

Support to the development of the Second Clean Air Outlook

Specific Contract 6 under Framework Contract ENV.C.3/FRA/2017/0012

Final Report

Markus Amann¹, Jens Borken-Kleefeld¹, Janusz Cofala¹, Chris Heyes¹, Lena Hoglund-Isaksson¹,
Gregor Kiesewetter¹, Zbigniew Klimont¹, Peter Rafaj¹, Wolfgang Schöpp¹, Fabian Wagner¹,
Wilfried Winiwarter¹, Mike Holland², Toon Vandyck³

¹ International Institute for Applied Systems Analysis (IIASA), Austria

² EMRC, UK

³ Joint Research Centre, European Commission, Spain

Contracting institutions

European Commission

Directorate General Environment

DG ENV.Unit A5
B-1049 Brussels
Belgium

International Institute for Applied Systems Analysis (IIASA)

Schlossplatz 1
A-2361 Laxenburg
Austria

Contact person:

Markus Amann
Phone: +43 2236 807 432
e-mail: amann@iiasa.ac.at

Laxenburg, December 21, 2020

Abstract

To support the development of the Second Clean Air Outlook of the European Union, this report presents a set of model calculations that explore the likely future development of air quality in Europe. The analysis incorporates recent methodological updates as well as the latest information on emission inventories, and projections and pollution control policies that have been reported by the Member States in their official inventory submissions and the National Air Pollution Control Programs.

The resulting emission projections are compared with the obligations of the National Emission reduction Commitments (NEC) Directive, and their impacts on future air quality are assessed. The report also examines how additional clean air and climate policies could alter the results of the baseline projection.

For each Member State, the report quantifies the contributions of other countries to PM_{2.5} exposure domestically, as well as the destinations of exported pollution. It is found that internationally coordinated policy action contributed a major share to air quality improvements in the past, and implementation of the NEC Directive will lead to further effective improvements in European air quality.

The model analysis conducted for this report clearly reveals net benefits from additional actions towards cleaner air, with positive macro-economic implications.

Disclaimer

This document has been prepared for the European Commission however it reflects the views only of the authors, and the Commission cannot be held responsible for any use which may be made of the information contained therein.

Table of Contents

1	Introduction.....	5
2	Policy scenarios considered in this report.....	7
2.1	Activity projections	7
2.1.1	The CAO2 baseline projection	7
2.1.2	The 1.5 TECH scenario	7
2.1.3	The 1.5 LIFE scenario	7
2.2	Air pollution control policies and measures	8
2.2.1	The CAO2 baseline emission control legislation	8
2.2.2	Additional national policies and measures	8
2.2.3	Maximum technically feasible emission reductions	8
2.3	Policy scenarios	8
3	Scenario results	10
3.1	Compliance with the NEC Directive	10
3.2	Air quality management zones: Background concentrations, EU Air Quality Standards and WHO guideline levels.....	16
3.2.1	Background concentrations of PM _{2.5}	16
3.2.2	Background concentrations of NO ₂	20
3.3	Health impact indicators.....	24
3.3.1	Indicators for exposure to PM _{2.5}	24
3.3.2	Indicators for exposure to ground-level ozone.....	29
3.4	Ecosystems impact indicators	31
3.5	Economic valuation and economic analysis.....	39
3.5.1	Valuation methods	39
3.5.2	Health benefits	42
3.5.3	Reduced damages to materials, crops, forests and ecosystems.....	42
3.5.4	Comparison of costs and benefits for health and non-health effects at the EU27 level	44
3.5.5	Macro-economic impacts.....	47
3.6	Results for the recent Mix55 climate policy scenario.....	53
3.7	Cross-border effects of national policies and measures.....	54
3.7.1	Import-export budgets for PM _{2.5}	54
3.7.2	Origin of PM _{2.5} in each Member State.....	59
3.7.3	Air quality benefits of emission controls on other countries.....	67
4	Key findings	68

List of acronyms

BC	Black carbon
CLE	Current legislation
CAO	Clean Air Outlook
CH ₄	Methane
CO ₂	Carbon dioxide
EEA	European Environment Agency
EMEP	European Monitoring and Evaluation Programme of the Convention on Long-range Transboundary Air Pollution
ERC	Emission Reduction Commitment of the NEC Directive
EU	European Union
GAINS	Greenhouse gas - Air pollution Interactions and Synergies model
GDP	Gross domestic product
GNFR	'Nomenclature For Reporting' of emission inventories to EMEP/EEA
IED	Industrial Emissions Directive
IIASA	International Institute for Applied Systems Analysis
IIR	Informative Inventory Reports
kt	kilotons = 10 ³ tons
MCP	Medium Sized Combustion Plants
MTFR	Maximum technically feasible emission reductions
NAPCP	National Air Pollution Control Programme
NEC	National Emission Reduction Commitments
NECD	National Emission Reduction Commitments Directive
NH ₃	Ammonia
NMVOC	Non-methane volatile organic compounds
NO _x	Nitrogen oxides
NO ₂	Nitrogen dioxide
NRMM	Non-road Mobile Machinery
O ₃	Ozone
PaMs	Policies and Measures
PJ	Petajoule = 10 ¹⁵ joule
PM ₁₀	Fine particles with an aerodynamic diameter of less than 10 µm
PM _{2.5}	Fine particles with an aerodynamic diameter of less than 2.5 µm
PRIMES	Energy Systems Model of the National Technical University of Athens
SO ₂	Sulphur dioxide
VOC	Volatile organic compounds
WAM	With Additional Measures (scenario)

1 Introduction

The 2013 Clean Air Programme proposed a regular report on the air quality situation in Europe¹, also covering the prospects for future emission reduction and progress towards the EU's objectives. The analytical basis for such a prospective exercise follows the work done in support to the preparation of Directive (EU) 2016/2284 on the reduction of national emissions of certain atmospheric pollutants² – the National Emission reduction Commitment Directive (NEC Directive). Along these lines, extensive model analysis (Amann, 2018) has supported the first Clean Air Outlook (COM(2018)446)³ in 2018.

This report summarizes work that has been conducted to inform the development of the Second Clean Air Outlook in 2020, addressing the following objectives:

1. Update the analytical framework for the analysis of policy scenarios,
2. incorporate latest data on emission inventories,
3. update the model databases to introduce policy developments since the adoption of the first Clean Air Outlook, and to include new information provided by Member States in the National Air Pollution Control Programmes (NAPCPs) and the National Energy and Climate Plans (NECPs),
4. develop policy scenarios to quantify:
 - the progress towards compliance with the air pollution reduction commitments under the NEC Directive and towards the achievement of EU environment and health objectives as set out in the 7th Environmental Action Program,
 - the related socio-economic impacts, and
 - further EU-level measures that would be needed to meet those objectives, and their expected impacts.

The main part of this report summarizes the key results of the project, while methodological details and comparisons with national statistics are provided in the Annex. In this main report, Section 2 describes the policy scenarios whose results are then analysed in the remainder of the report. Section 3.1 discusses how the alternative policy assumptions affect emissions of the key air pollutants, and in particular the achievement of the emission reduction requirements of the NEC Directive. Section 3.2 assesses the impacts of the emission changes on air quality, addressing the implications for compliance with the air quality limit values set in the Ambient Air Quality Directives⁴ for PM_{2.5} and NO₂ as well as for the achievement of the WHO air quality guideline value for PM_{2.5}. Consequences for health impact indicators are summarised in Section 3.3, and for ecosystems impacts in Section 3.4. Section 3.5 presents estimates of the monetised benefits of health, ecosystems and crop impacts, and provides a full macro-economic assessment of the costs and benefits of the scenarios. While the main policy scenarios of this report have been developed for the state of climate policies in 2019, implications of the latest climate policy proposal from the European Commission in the context of the EU Green Deal are discussed in Section 3.6. The importance of international cooperation for achieving clean air, and the benefits from EU-wide

¹ This comes in complement to the ex post assessments produced every year by the European Environment Agency (EEA).

² Directive (EU) 2016/2284 of the European Parliament and of the Council of 14 December 2016 on the reduction of national emissions of certain atmospheric pollutants, amending Directive 2003/35/EC and repealing Directive 2001/81/EC. OJ L 344, 17.12.2016.

³ https://ec.europa.eu/environment/air/clean_air/outlook.htm

⁴ Directive 2008/50/EC of the European Parliament and of the Council of 21 May 2008 on ambient air quality and cleaner air for Europe. OJ L 152, 11.6.2008.

coordinated air quality management, are presented in Section 3.7. Key findings are summarised in Section 4.

2 Policy scenarios considered in this report

To explore the prospects for future emissions and air quality in the EU, this report explores seven policy scenarios that are based on different combinations of activity projections for different climate policy choices and air pollution control legislation (Table 2.1).

2.1 Activity projections

2.1.1 The CAO2 baseline projection

The CAO2 baseline activity projection employs for the period up to 2030 the 'EUCO 32 32.5' scenario developed with the PRIMES model to support the Commission's June 2019 assessment of the draft national energy and climate plans (NECPs) of the Member States. While it maintains the assumptions on macro-economic development and fossil fuels prices of the Reference scenario 2016 that has been employed for the first Clean Air Outlook, the scenario assumes achievement of the EU energy efficiency target of 32.5% and a renewable energy target of 32% as agreed in the 'Clean energy for all Europeans' package until 2030, and implementation of the current policies on non-CO₂ greenhouse gas emissions⁵. Beyond 2030, the CAO2 baseline follows the PRIMES baseline projections that extend up to 2050. For the purposes of air quality assessment and in order to align with the energy statistics that have been used by Member States for their emission inventories, additional structural detail has been added to the original PRIMES projections.

To explore the potential implications of additional climate policies, two decarbonisation scenarios of the EU energy and agricultural systems in line with a stabilisation of global temperature increase at 1.5°C are analysed.

2.1.2 The 1.5 TECH scenario

The 1.5 TECH scenario explores technological options, including electrification of all end-uses, of hydrogen as an energy carrier, emergence of carbon-neutral hydrocarbons and efficiency savings, in order to achieve net zero GHG emissions by 2050. Remaining emissions that cannot be abated by 2050 are balanced by negative emissions through the deployment of biomass associated with carbon capture and storage (BECCS) and LULUCF sinks.

2.1.3 The 1.5 LIFE scenario

In contrast, the 1.5 LIFE scenario explores the transformation of EU business and consumption patterns towards a more circular economy. Increased climate awareness of EU citizens translates into lifestyle changes and consumer choices that are more beneficial for climate. These include a continued trend of EU consumers towards less carbon intensive diets, a sharing economy in transport, limiting growth in air transport demand and a more rational use of energy demand for heating and cooling. While both climate policy scenarios include additional incentives to enhance LULUCF sinks, such incentives are much larger in the 1.5 LIFE scenario.

For the development of the agricultural sector in the 1.5 LIFE scenario, dietary shifts have been assumed that reduce total animal calorie intake levels to the recommended levels while keeping the total calorie consumption constant. The declining consumption trends of bovine, sheep and goat meat and milk observed in the EU in the last decade are assumed to further continue. Consumption of pig and poultry meat as well as eggs would remain at the level of 2010.

⁵ https://ec.europa.eu/clima/sites/clima/files/docs/pages/com_2018_733_analysis_in_support_en_0.pdf

2.2 Air pollution control policies and measures

Air pollution control policies and measures that are adopted in the various Member States constitute the second important component of the policy scenarios presented in this report. The following sets of policies and measures have been compiled

2.2.1 The CAO2 baseline emission control legislation

This package includes the EU legislation on air pollution controls as well as already implemented national policies and measures. For each of these regulations, the GAINS database holds for each affected source category in each Member State the temporal penetration, i.e., the evolution over time of the shares of the emission sources that are subject to the specific regulation.

2.2.2 Additional national policies and measures

This package includes additional national policies and measures that have been reported by Member States in their NAPCPs and PaMs databases as 'selected for adoption' (see Annex 2.2.2), provided they were described with sufficient level of detail, in addition to the policies and measures of the CAO2 baseline legislation.

2.2.3 Maximum technically feasible emission reductions

The third variant outlines the scope of emission reductions that could be achieved through full application of all technical emission control measures, as represented in the GAINS database. While excluding premature scrapping of existing capital stock, this scenario assumes that all new (expansion and replacement) investments would employ the most efficient emission control measures that are currently available, subject to constraints emerging from country-specific conditions (e.g., structural composition of emission sources, vintage distribution, climatic and topographic conditions, firm and farm size distributions, etc.). As a conservative assumption, no further technological progress is assumed in this scenario.

2.3 Policy scenarios

This report examines future air quality for seven alternative combinations of activity projections and emission control packages (Table 2.1).

As a benchmark, the CAO2 baseline combines the CAO2 baseline activity projection with the baseline air pollution control legislation, which includes already adopted EU-wide and national legislation. The NAPCP scenario explores the impacts of the additional policies and measures that are reported in the NAPCPs as 'selected for adoption'. Potential side effects of tightened climate policy measures on air quality are explored through three climate policy scenarios, i.e., the 1.5 TECH scenario, the 1.5 LIFE scenario, and the recent Mix55 scenario that has been used in the impact assessment supporting the Commission proposal for reducing GHG by 55% by 2030⁶. These climate scenarios are combined with clean air scenarios representing respectively the NAPCP air pollution control measures and the additional improvements that could be achieved through the maximum feasible application of all available technical air pollutant emission control measures.

⁶ Communication from The Commission to The European Parliament, The Council, The European Economic and Social Committee and The Committee Of The Regions: Stepping up Europe's 2030 climate ambition Investing in a climate-neutral future for the benefit of our people. COM/2020/562 final

Table 2.1: Policy scenarios considered in this report

Scenario name	Climate policies	Air pollution control measures	Years
Baseline scenarios			
CAO2 baseline	EUCO 32-32.5.5 up to 2030; afterwards baseline of the EU 2050 climate strategy vision for the 2050 horizon	Latest EU-wide legislation and already adopted national measures	2005, 2010, 2015, 2020, 2025, 2030, 2035, 2040, 2045, 2050
NAPCP	As above	As above + additional measures (PaMs selected for adoption) of the NAPCPs	2025, 2030, 2035, 2040, 2045, 2050
Climate policy scenarios			
1.5 TECH+WAM	1.5 TECH scenario of the EU 2050 climate strategy vision	With the additional measures as in the NAPCP scenario	2035, 2040, 2045, 2050
1.5 LIFE+WAM	1.5 LIFE scenario of the EU 2050 climate strategy vision, with additional assumptions on diet changes	With the additional measures as in the NAPCP scenario	2035, 2040, 2045, 2050
Air pollution control scenarios			
CAO2 baseline+MTFR	EUCO 32 32.5 until 2030 and thereafter the baseline of the EU 2050 climate strategy vision	With maximum technically feasible emission reduction (MTFR) measures	2030, 2050
1.5 LIFE+MTFR	1.5 LIFE scenario of the EU 2050 climate strategy vision, with additional assumptions on diet changes	With maximum technically feasible emission reduction (MTFR) measures	2050
The latest climate policy scenario			
Mix 55+WAM	The Mix55 scenario from the 2020 Commission proposal ⁷	With the additional measures as in the NAPCP scenario	2030, 2040, 2050

⁷ Communication From The Commission To The European Parliament, The Council, The European Economic And Social Committee And The Committee Of The Regions: Stepping up Europe's 2030 climate ambition Investing in a climate-neutral future for the benefit of our people. COM/2020/562 final

3 Scenario results

For each of the seven policy scenarios listed in Table 2.1, emissions, emission control costs, concentration maps of the relevant air pollutants, as well as indicators for health and ecosystems impacts have been calculated, with results reported for each Member State and key economic sector.

3.1 Compliance with the NEC Directive

Table 3.1 to Table 3.5 compare, for each Member State, the emission reduction commitments of the NEC Directive for 2020-29 and 2030 onwards with the emission projections of the baseline scenario, considering the most recent updates to the emission inventories for 2005.

For 2020, the CAO2 baseline projection suggests for several Member States higher emissions than what is required in the NEC Directive. The baseline projections for SO₂ do not meet the NEC Directive requirements for one country, for NO_x for four countries, for NH₃ for 10 countries, and for NMVOC for two Member States. However, it has to be noted that the activity statistics that are used in the CAO2 baseline projection for 2020 do not reflect actual statistics as this report has been developed before the end of 2020. Thus, they do not consider any impacts of the COVID-19 crisis. Instead, they are derived from projections for 2020 that have been developed in 2018 for the EUCO 32 32.5 scenario.

For 2030, the CAO2 baselines for the energy-related pollutants SO₂, NO_x, PM_{2.5} and NMVOC achieve the emission reduction commitments (ERC) of the NEC Directive for most Member States. For SO₂, emissions are slightly higher in Slovenia; for NO_x, higher emissions are computed for the Czech Republic and Luxembourg, and for PM_{2.5} for Germany and the Netherlands. For NMVOC, the CAO2 baseline would require further measures in Ireland and Luxembourg.

In contrast, there are major gaps between the NH₃ emissions of the CAO2 baseline and the 2030 ERCs in 22 out of the 27 Member States. For seven Member States the additional policies and measures that have been reported in the NAPCPs as 'selected for adoption' would align NH₃ emissions with the ERCs. This would however mean that 15 Member States would not comply with their NEC Directive obligations.

Table 3.1: Comparison of the baseline scenarios with the emission reduction commitments (ERCs) of the NEC Directive for SO₂ emissions. Highlighted cells indicate non-achievement of the ERCs.

	2005	2020				2030					
		ERC	CAO2 baseline	NECD threshold	CAO2 baseline	ERC	CAO2 baseline	NAPCP	NECD threshold	CAO2 baseline	NAPCP
		% reduction compared to 2005 levels			kilotons	% reduction compared to 2005 levels				kilotons	
Austria	26.2	26%	50%	19.4	13.1	41%	62%	62%	15.5	9.9	9.9
Belgium	141.1	43%	73%	80.4	38.0	66%	79%	83%	48.0	29.5	24.0
Bulgaria	758.3	78%	91%	166.8	69.6	88%	94%	94%	91.0	48.2	48.2
Croatia	56.8	55%	79%	25.5	11.9	83%	87%	87%	9.6	7.3	7.3
Cyprus	36.4	83%	94%	6.2	2.2	93%	95%	95%	2.6	1.7	1.7
Czech Rep.	212.5	45%	68%	116.9	67.0	66%	77%	77%	72.2	49.1	49.1
Denmark	26.0	35%	59%	16.9	10.8	59%	72%	72%	10.7	7.4	7.4
Estonia	75.8	32%	66%	51.6	25.5	68%	73%	73%	24.3	20.4	20.4
Finland	71.0	30%	52%	49.7	34.1	34%	66%	66%	46.8	23.9	23.9
France	431.7	55%	67%	194.3	141.5	77%	79%	79%	99.3	90.0	89.1
Germany	478.2	21%	33%	377.8	321.6	58%	58%	58%	200.8	199.6	199.6
Greece	542.1	74%	86%	140.9	76.6	88%	94%	94%	65.0	34.6	34.6
Hungary	41.6	46%	70%	22.5	12.5	73%	84%	84%	11.2	6.6	6.6
Ireland	72.0	65%	81%	25.2	13.5	85%	90%	90%	10.8	7.1	7.1
Italy	386.0	35%	71%	250.9	113.4	71%	77%	77%	111.9	89.3	89.3
Latvia	8.5	8%	57%	7.8	3.7	46%	62%	62%	4.6	3.2	3.2
Lithuania	29.3	55%	60%	13.2	11.6	60%	69%	70%	11.7	9.1	8.9
Luxembourg	2.3	34%	44%	1.5	1.3	50%	55%	55%	1.2	1.0	1.0
Malta	12.1	77%	96%	2.8	0.5	95%	96%	96%	0.6	0.5	0.5
Netherlands	68.2	28%	62%	49.1	25.9	53%	65%	65%	32.0	23.5	23.5
Poland	1181.9	59%	68%	484.6	379.5	70%	82%	82%	354.6	208.2	208.2
Portugal	190.0	63%	86%	70.3	26.8	83%	88%	88%	32.3	22.8	22.8
Romania	599.6	77%	89%	137.9	67.9	88%	92%	92%	72.0	47.5	47.5
Slovakia	88.3	57%	78%	38.0	19.1	82%	85%	85%	15.9	13.3	13.3
Slovenia	40.2	63%	87%	14.9	5.4	92%	92%	92%	3.2	3.4	3.4
Spain	1189.8	67%	86%	392.6	168.6	88%	93%	93%	142.8	80.2	79.8
Sweden	37.1	22%	57%	28.9	15.8	22%	62%	62%	28.9	14.2	14.2
EU-27	6803.0	59%	75%	2786.5	1677.1	78%	85%	85%	1519.5	1051.6	1044.5

Table 3.2: Comparison of the baseline scenarios with the emission reduction commitments (ERCs) of the NEC Directive for NO_x emissions. Highlighted cells indicate non-achievement of the ERCs.

	2005	2020				2030					
		ERC	CAO2 baseline	NECD threshold	CAO2 baseline	ERC	CAO2 baseline	NAPCP	NECD threshold	CAO2 baseline	NAPCP
		% reduction compared to 2005 levels			kilotons	% reduction compared to 2005 levels				kilotons	
Austria	233.5	37%	49%	147.1	119.7	69%	77%	77%	72.4	53.1	52.6
Belgium	292.7	41%	52%	172.7	140.3	59%	71%	72%	120.0	84.8	80.6
Bulgaria	171.6	41%	48%	101.3	88.7	58%	64%	64%	72.1	62.3	62.3
Croatia	75.6	31%	48%	52.2	39.2	57%	72%	72%	32.5	21.4	21.4
Cyprus	22.9	44%	65%	12.8	8.0	55%	77%	77%	10.3	5.2	5.2
Czech Rep.	262.0	35%	40%	170.3	156.7	64%	59%	59%	94.3	108.6	107.7
Denmark	187.3	56%	56%	82.4	82.2	68%	75%	75%	59.9	46.6	46.4
Estonia	39.0	18%	36%	32.0	24.9	30%	56%	56%	27.3	17.3	17.3
Finland	187.1	35%	41%	121.6	110.3	47%	58%	58%	99.2	79.2	79.2
France	1347.5	50%	47%	673.8	719.6	69%	76%	76%	417.7	326.2	326.2
Germany	1423.7	39%	38%	868.4	876.2	65%	66%	67%	498.3	480.8	469.4
Greece	428.7	31%	62%	295.8	161.9	55%	79%	79%	192.9	88.7	88.7
Hungary	155.7	34%	50%	102.8	77.7	66%	71%	71%	52.9	44.7	44.7
Ireland	139.4	49%	51%	71.1	68.5	69%	74%	73%	43.2	36.3	37.6
Italy	1200.2	40%	48%	720.1	618.2	65%	71%	71%	420.1	342.9	342.9
Latvia	40.4	32%	21%	27.5	32.1	34%	45%	45%	26.7	22.4	22.4
Lithuania	49.3	48%	31%	25.6	34.1	51%	58%	62%	24.1	20.6	18.8
Luxembourg	55.3	43%	57%	31.5	23.7	83%	83%	83%	9.4	9.6	9.6
Malta	9.9	42%	77%	5.8	2.3	79%	83%	83%	2.1	1.6	1.6
Netherlands	356.1	45%	46%	195.8	192.3	61%	67%	67%	138.9	118.2	118.2
Poland	771.7	30%	33%	540.2	516.0	39%	56%	56%	470.7	341.4	341.4
Portugal	279.7	36%	54%	179.0	128.2	63%	71%	71%	103.5	81.7	81.7
Romania	286.9	45%	46%	157.8	154.6	60%	64%	64%	114.8	104.6	104.6
Slovakia	99.3	36%	47%	63.6	52.7	50%	63%	63%	49.7	37.0	37.0
Slovenia	50.2	39%	46%	30.6	27.0	65%	69%	69%	17.6	15.7	15.7
Spain	1306.0	41%	54%	770.5	599.3	62%	75%	75%	496.3	328.5	327.5
Sweden	201.7	36%	52%	129.1	96.0	66%	76%	79%	68.6	49.1	42.1
EU-27	9673.3	40%	47%	5781.3	5150.7	61%	70%	70%	3735.4	2928.6	2902.9

Table 3.3: Comparison of the baseline scenarios with the emission reduction commitments (ERCs) of the NEC Directive for PM_{2.5} emissions. Highlighted cells indicate non-achievement of the ERCs.

	2005	2020				2030					
		ERC	CAO2 baseline	NECD threshold	CAO2 baseline	ERC	CAO2 baseline	NAPCP	NECD threshold	CAO2 baseline	NAPCP
		% reduction compared to 2005 levels			kilotons	% reduction compared to 2005 levels				kilotons	
Austria	21.8	20%	33%	17.4	14.6	46%	56%	56%	11.8	9.6	9.5
Belgium	33.5	20%	13%	26.8	29.2	39%	41%	43%	20.5	19.9	19.2
Bulgaria	38.3	20%	15%	30.7	32.6	41%	63%	71%	22.6	14.1	11.3
Croatia	41.6	18%	76%	34.1	9.9	55%	87%	87%	18.7	5.4	5.4
Cyprus	2.4	46%	64%	1.3	0.9	70%	72%	72%	0.7	0.7	0.7
Czech Rep.	45.6	17%	46%	37.8	24.6	60%	69%	69%	18.2	14.4	14.1
Denmark	26.8	33%	30%	18.0	18.8	55%	65%	68%	12.1	9.4	8.5
Estonia	14.0	15%	44%	11.9	7.8	41%	72%	72%	8.3	4.0	4.0
Finland	28.5	30%	29%	19.9	20.2	34%	49%	49%	18.8	14.5	14.5
France	261.1	27%	25%	190.6	195.4	57%	61%	63%	112.3	101.7	97.4
Germany	129.1	26%	28%	95.5	93.4	43%	42%	42%	73.6	75.5	75.4
Greece	53.5	35%	49%	34.8	27.3	50%	70%	70%	26.7	15.8	15.8
Hungary	42.5	13%	31%	37.0	29.4	55%	66%	68%	19.1	14.3	13.5
Ireland	19.9	18%	33%	16.3	13.3	41%	59%	55%	11.7	8.1	9.0
Italy	166.9	10%	12%	150.2	146.0	40%	55%	58%	100.1	75.5	70.3
Latvia	23.2	16%	38%	19.5	14.3	43%	69%	72%	13.2	7.3	6.5
Lithuania	19.2	20%	31%	15.3	13.2	36%	74%	76%	12.3	4.9	4.6
Luxembourg	2.8	15%	28%	2.4	2.0	40%	41%	41%	1.7	1.6	1.6
Malta	0.7	25%	75%	0.5	0.2	50%	79%	79%	0.4	0.2	0.2
Netherlands	23.4	37%	29%	14.8	16.5	45%	42%	42%	12.9	13.6	13.6
Poland	252.6	16%	16%	212.2	211.1	58%	58%	58%	106.1	105.3	105.3
Portugal	64.9	15%	39%	55.2	39.9	53%	60%	60%	30.5	26.1	26.1
Romania	123.0	28%	14%	88.6	105.6	58%	69%	69%	51.7	37.9	37.9
Slovakia	34.8	36%	77%	22.3	8.0	49%	84%	84%	17.7	5.7	5.7
Slovenia	13.2	25%	29%	9.9	9.3	60%	63%	66%	5.3	4.9	4.4
Spain	147.2	15%	19%	125.1	119.7	50%	61%	64%	73.6	57.8	53.0
Sweden	34.3	19%	41%	27.8	20.1	19%	54%	54%	27.8	15.9	15.9
EU-270	1664.8	21%	27%	1315.8	1222.9	50%	60%	61%	828.3	664.0	643.3

Table 3.4: Comparison of the baseline scenarios with the emission reduction commitments (ERCs) of the NEC Directive for NH₃ emissions. Highlighted cells indicate non-achievement of the ERCs. Negative numbers indicate an increase in emissions.

	2005	2020				2030					
		ERC	CAO2 baseline	NECD threshold	CAO2 baseline	ERC	CAO2 baseline	NAPCP	NECD threshold	CAO2 baseline	NAPCP
		% reduction compared to 2005 levels			kilotons	% reduction compared to 2005 levels				kilotons	
Austria	65.3	1%	-8%	64.7	70.8	12%	-11%	5%	57.5	72.6	62.1
Belgium	72.0	2%	3%	70.5	69.8	13%	3%	17%	62.6	69.8	59.7
Bulgaria	46.7	3%	0%	45.3	46.8	12%	0%	6%	41.1	47.0	43.8
Croatia	47.6	1%	13%	47.1	41.3	25%	12%	31%	35.7	41.7	32.9
Cyprus	7.0	10%	11%	6.3	6.2	20%	11%	11%	5.6	6.2	6.2
Czech Rep.	75.5	7%	14%	70.2	65.0	22%	20%	30%	58.9	60.7	52.6
Denmark	76.6	24%	17%	58.2	63.5	24%	17%	17%	58.2	63.3	63.3
Estonia	9.7	1%	-27%	9.6	12.4	1%	-28%	-12%	9.6	12.5	10.9
Finland	37.6	20%	16%	30.1	31.6	20%	13%	13%	30.1	32.6	32.6
France	621.7	4%	4%	596.9	596.0	13%	6%	21%	540.9	585.0	493.9
Germany	627.2	5%	5%	595.9	593.2	29%	11%	29%	445.3	559.0	446.0
Greece	73.0	7%	13%	67.9	63.5	10%	15%	15%	65.7	61.8	61.8
Hungary	79.0	10%	15%	71.1	67.0	32%	21%	35%	53.7	62.5	51.5
Ireland	103.0	1%	-6%	101.9	109.3	5%	-11%	-11%	97.8	114.2	114.2
Italy	433.0	5%	12%	411.3	379.9	16%	14%	19%	363.7	372.0	350.5
Latvia	15.3	1%	-16%	15.2	17.7	1%	-15%	-5%	15.2	17.6	16.1
Lithuania	33.1	10%	3%	29.8	32.2	10%	0%	0%	29.8	33.0	33.0
Luxembourg	5.7	1%	2%	5.7	5.6	22%	-1%	-1%	4.5	5.7	5.7
Malta	1.5	4%	24%	1.4	1.1	24%	27%	27%	1.1	1.1	1.1
Netherlands	151.7	13%	21%	132.0	120.3	21%	24%	24%	119.9	115.9	115.9
Poland	325.0	1%	-1%	321.8	327.5	17%	-5%	10%	269.8	340.8	291.4
Portugal	54.2	7%	8%	50.4	49.7	15%	5%	5%	46.1	51.3	51.3
Romania	189.5	13%	17%	164.8	157.9	25%	19%	19%	142.1	152.6	152.6
Slovakia	38.8	15%	31%	33.0	26.8	30%	32%	32%	27.2	26.5	26.5
Slovenia	20.2	1%	16%	20.0	17.0	15%	18%	18%	17.1	16.5	16.5
Spain	520.5	3%	4%	504.9	498.5	16%	4%	21%	437.2	499.5	412.2
Sweden	59.5	15%	14%	50.6	51.2	17%	15%	16%	49.4	50.7	49.8
EU-270	3790.0	6%	7%	3576.5	3521.6	19%	8%	19%	3085.8	3472.0	3054.1

Table 3.5: Comparison of the baseline scenarios with the emission reduction commitments (ERCs) of the NEC Directive for NMVOC emissions. Highlighted cells indicate non-achievement of the ERCs.

	2005	2020				2030					
		ERC	CAO2 baseline	NECD threshold	CAO2 baseline	ERC	CAO2 baseline	NAPCP	NECD threshold	CAO2 baseline	NAPCP
		% reduction compared to 2005 levels				% reduction compared to 2005 levels					
					kilotons					kilotons	
Austria	122.8	21%	34%	97.0	80.9	36%	47%	47%	78.6	64.6	64.6
Belgium	141.9	21%	37%	112.1	88.9	35%	44%	45%	92.2	79.1	78.1
Bulgaria	96.7	21%	32%	76.4	65.7	42%	54%	57%	56.1	44.8	41.2
Croatia	104.8	34%	62%	69.2	40.2	48%	68%	68%	54.5	33.8	33.8
Cyprus	13.8	45%	50%	7.6	6.9	50%	60%	60%	6.9	5.5	5.5
Czech Rep.	261.5	18%	43%	214.4	148.8	50%	52%	52%	130.7	125.3	124.6
Denmark	117.3	35%	45%	76.3	64.9	37%	55%	56%	73.9	52.7	51.5
Estonia	27.8	10%	42%	25.0	16.3	28%	53%	53%	20.0	13.0	13.0
Finland	121.2	35%	48%	78.8	63.6	48%	57%	57%	63.0	52.5	52.5
France	1184.9	43%	47%	675.4	630.6	52%	58%	58%	568.7	500.4	496.3
Germany	1158.7	13%	31%	1008.1	800.0	28%	37%	37%	834.3	732.0	731.3
Greece	295.1	54%	58%	135.7	124.2	62%	67%	67%	112.1	97.5	97.5
Hungary	146.2	30%	48%	102.3	76.2	58%	60%	60%	61.4	58.8	57.9
Ireland	77.2	25%	18%	57.9	63.1	32%	29%	27%	52.5	54.7	56.3
Italy	1216.7	35%	38%	790.9	757.5	46%	48%	49%	657.0	628.5	625.8
Latvia	41.9	27%	38%	30.6	25.9	38%	46%	48%	26.0	22.5	22.0
Lithuania	49.2	32%	41%	33.5	29.0	47%	58%	58%	26.1	20.9	20.4
Luxembourg	12.0	29%	29%	8.5	8.5	42%	40%	40%	7.0	7.3	7.3
Malta	3.7	23%	44%	2.8	2.0	27%	48%	48%	2.7	1.9	1.9
Netherlands	193.1	8%	17%	177.6	159.8	15%	24%	24%	164.1	147.4	147.4
Poland	642.4	25%	22%	481.8	499.8	26%	44%	44%	475.3	360.8	360.8
Portugal	181.9	18%	38%	149.2	112.4	38%	49%	49%	112.8	92.4	92.4
Romania	296.3	25%	36%	222.2	188.3	45%	63%	63%	163.0	110.1	110.1
Slovakia	139.2	18%	64%	114.1	50.7	32%	65%	65%	94.7	48.8	48.8
Slovenia	39.1	23%	44%	30.1	22.0	53%	54%	55%	18.4	17.9	17.4
Spain	701.6	22%	31%	547.3	483.1	39%	39%	40%	428.0	426.1	419.6
Sweden	178.6	25%	40%	133.9	106.6	36%	44%	44%	114.3	99.6	99.6
EU-270	7565.4	28%	38%	5458.6	4715.9	41%	48%	49%	4494.2	3898.8	3877.6

3.2 Air quality management zones: Background concentrations, EU Air Quality Standards and WHO guideline levels

To inform considerations about the implications of the various emission scenarios related to exceedances above EU air quality standards and WHO air quality guideline levels, background concentration values of PM_{2.5} and NO₂ have been computed with the GAINS model at a 7 km*7 km spatial resolution. (Calculations for PM₁₀ could not be performed for this report as these would require substantial methodological updates in the GAINS model⁸). Also, in contrast to earlier studies, this analysis refrains from estimating concentrations for individual monitoring sites, in particular since the GAINS methodology applied for this report has not yet been upgraded to incorporate recent progress in fine-scale scale modelling of these pollutants.

For exceedance analysis, background concentrations computed for the 7 km*7 km grid cells have been allocated to the air quality management zones, and for each zone the maximum computed grid value has been taken.

3.2.1 Background concentrations of PM_{2.5}

For PM_{2.5}, the distributions of the highest background values that occur in the various air quality management zones in key scenarios are summarised in Table 3.6. Results for individual Member States are presented in Figure 3.1 to Figure 3.4.

Table 3.6: Distribution of the computed maximum background levels of PM_{2.5} over the air quality management zones in the EU-27

Concentration range	2015	2030 Baseline	2030 Baseline+MTFR	2050 Baseline	2050 1.5 LIFE+MTFR
Number of air quality management zones					
<10 µg/m ³	277	604	663	650	674
10-15 µg/m ³	243	61	10	24	0
15-20 µg/m ³	95	8	1	0	0
20-25 µg/m ³	39	1	0	0	0
>25 µg/m ³	20	0	0	0	0

For all 20 zones for which in 2015 computed maximum background concentrations exceeded the annual PM_{2.5} value of 25 µg/m³, concentrations are estimated to fall below this level by 2030 even in the CAO2 baseline. The number of zones for which computed maximum background levels of PM_{2.5} are below the current WHO guideline value of 10 µg/m³ is expected to increase from 277 (41% of all zones) in 2015 to 604 (90%) in the 2030 CAO2 baseline, and to 650 (96%) in 2050. The 1.5 LIFE+MTFR scenario would bring background concentrations in all zones below the current WHO guideline value.

⁸ However, since PM_{2.5} is a subset of PM₁₀, mass based PM₁₀ estimates will always be higher than for PM_{2.5}.

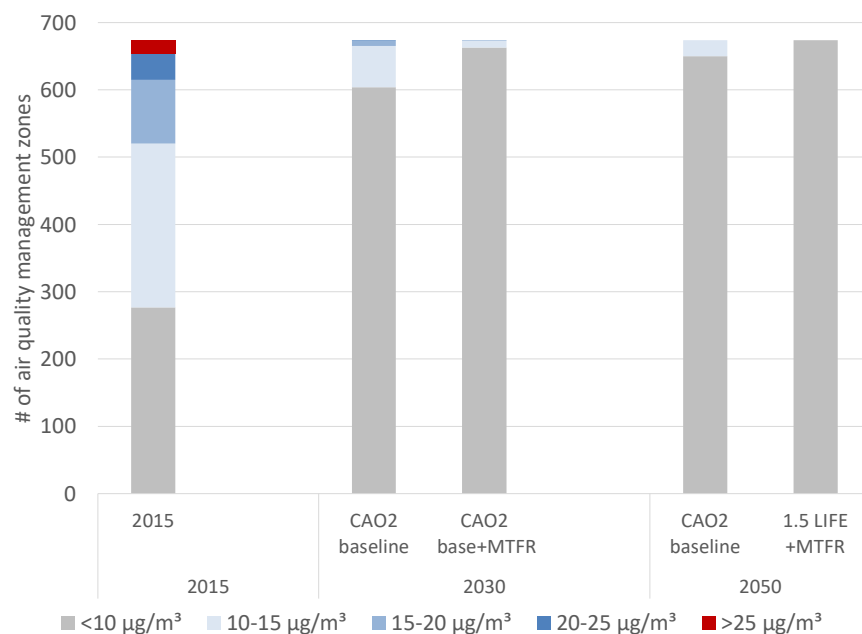


Figure 3.1: Distribution of air quality management zones according to their highest background concentrations of PM_{2.5}, EU-27

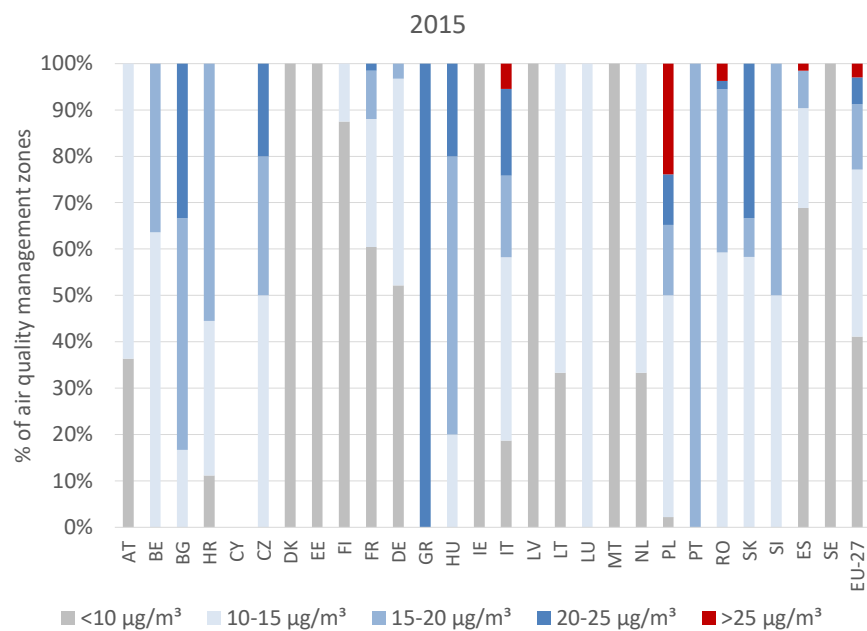


Figure 3.2: Distribution of air quality management zones according to their computed maximum level of background concentrations of PM_{2.5}, 2015. Cyprus is outside the geographical model domain. Note that the graph shows modelling results for background concentrations, which cannot be directly compared to monitoring data of total PM_{2.5}.

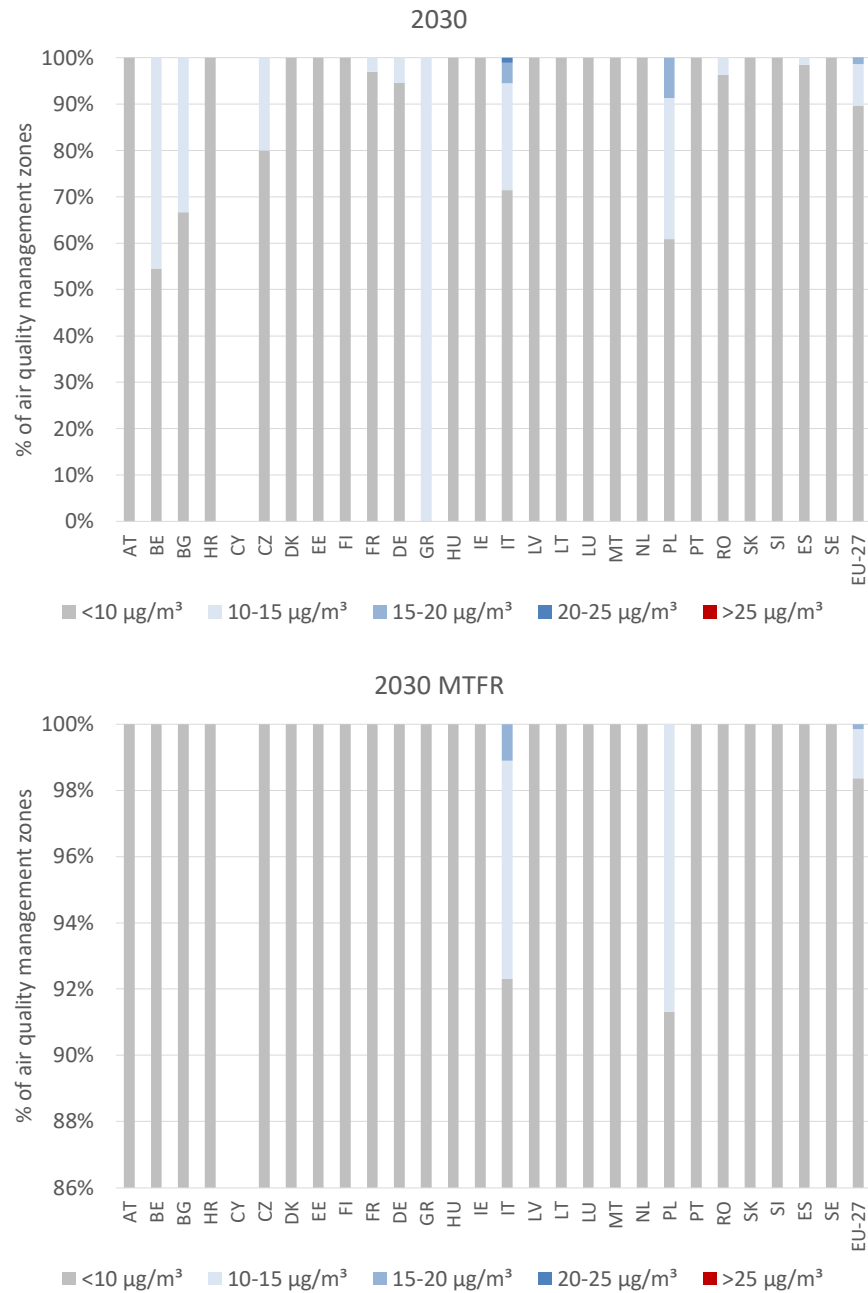


Figure 3.3: Distribution of air quality management zones according to their level of background concentrations of $PM_{2.5}$, 2030. Upper panel: CAO2 baseline scenario, lower panel: baseline with maximum technical air pollution controls (MTFR). Cyprus is outside the geographical model domain.

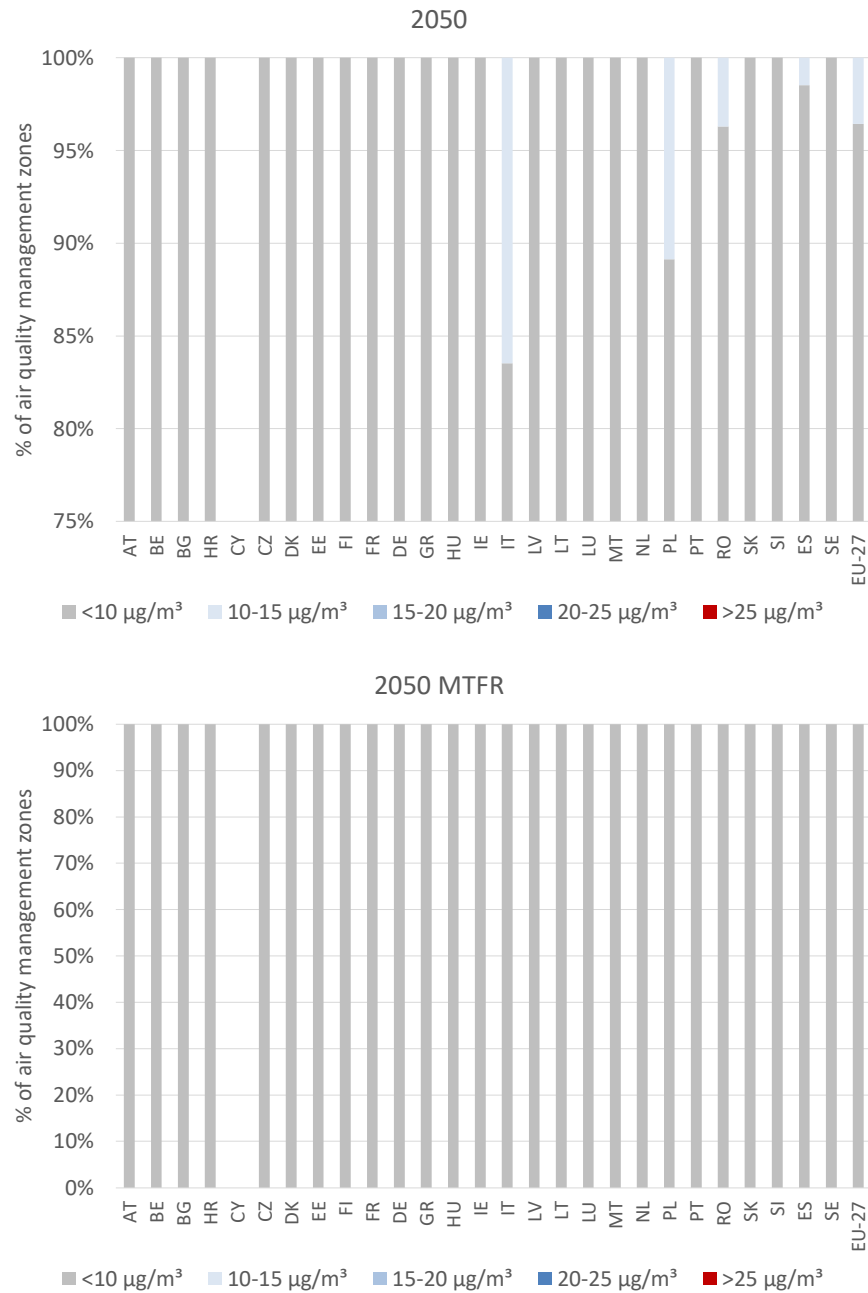


Figure 3.4: Distribution of air quality management zones according to their level of background concentrations of $\text{PM}_{2.5}$, 2050. Upper panel: CAO2 baseline scenario, lower panel: 1.5 LIFE scenario with maximum technical air pollution controls (MTRF). Cyprus is outside the geographical model domain.

3.2.2 Background concentrations of NO₂

For NO₂, the distributions of the computed maximum annual mean concentrations in each air quality management zone in the key scenarios are summarised in Table 3.7 and Figure 3.5. Results for individual Member States are presented in Figure 3.6 to Figure 3.8.

Table 3.7: Distribution of the computed highest background concentrations of NO₂ in the air quality management zones, EU-27

Concentration range	2015	2030 CAO2 baseline	2030 Baseline+MTFR	2050 Baseline	2050 1.5 LIFE+MTFR
Number of air quality management zones					
<10 µg/m ³	155	487	544	633	677
10-20 µg/m ³	269	175	142	67	28
20-30 µg/m ³	151	31	13	5	0
30-40 µg/m ³	56	9	6	0	0
>40 µg/m ³	74	3	0	0	0

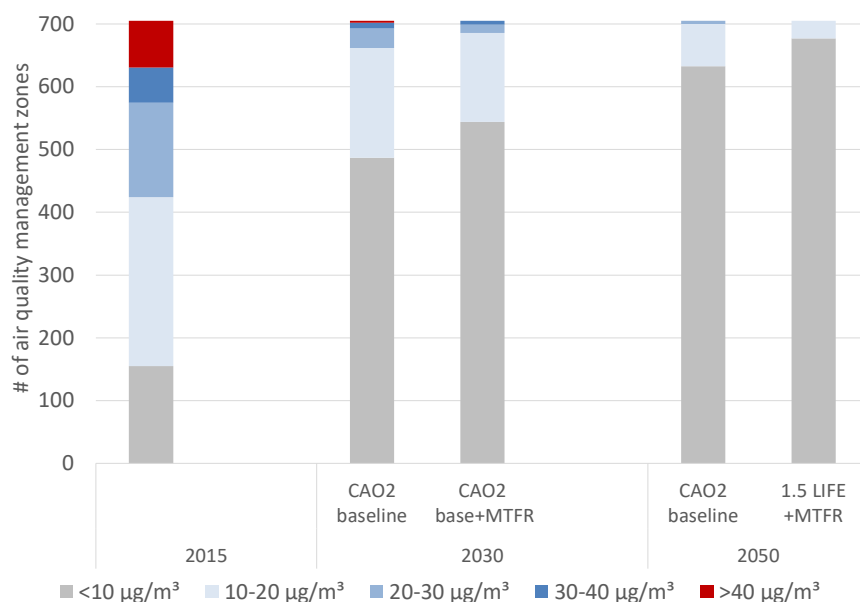


Figure 3.5: Distribution of air quality management zones according to their highest background concentrations of NO₂, EU-27

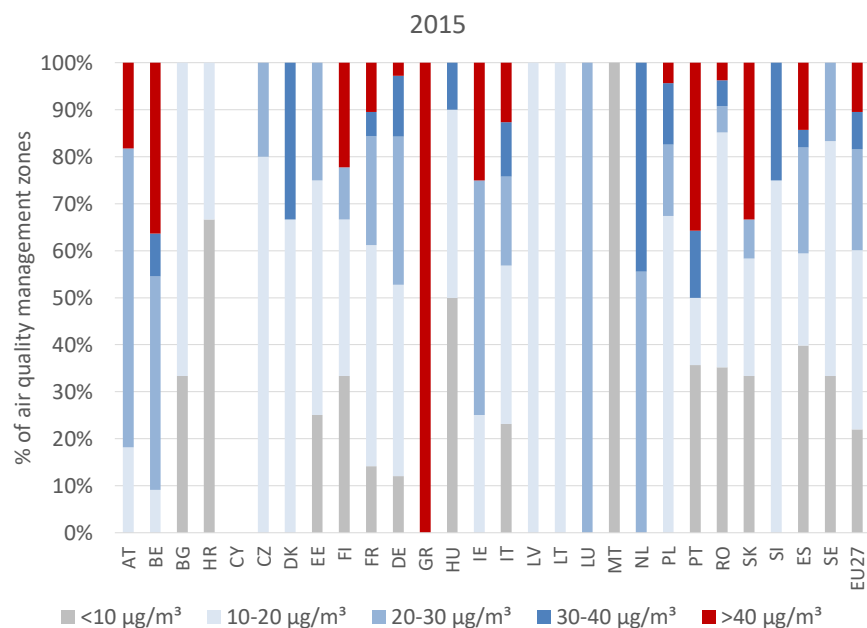


Figure 3.6: Distribution of computed maximum background concentrations of NO₂ in the air quality management zones, 2015. Cyprus is outside the geographical model domain. Note that the graph shows modelling results for background concentrations, which cannot be directly compared to monitoring data of total NO₂.

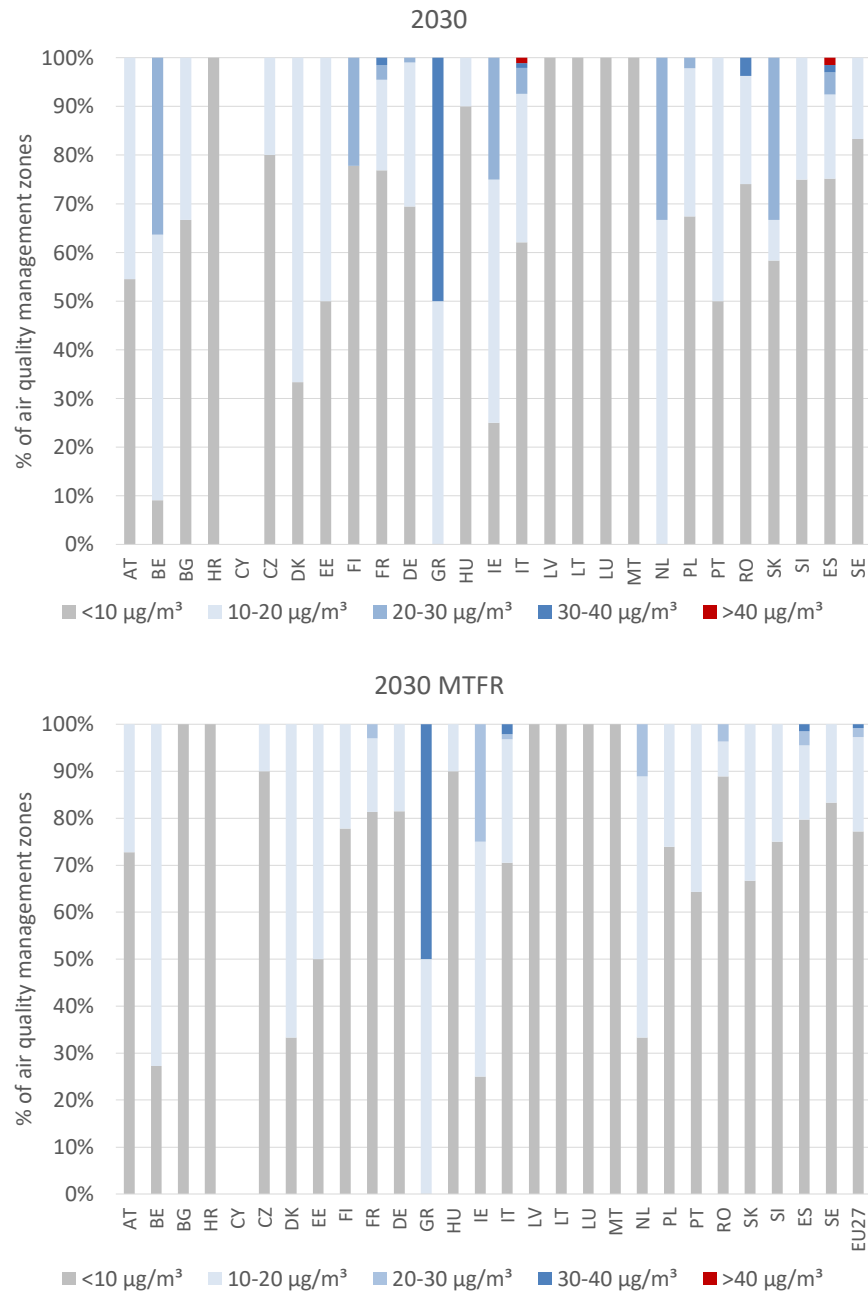


Figure 3.7: Distribution of maximum background concentrations of NO₂ in the air quality management zones, 2030. Upper panel: CAO2 baseline scenario, lower panel: CAO2 baseline with maximum technical air pollution controls. Cyprus is outside the geographical model domain.

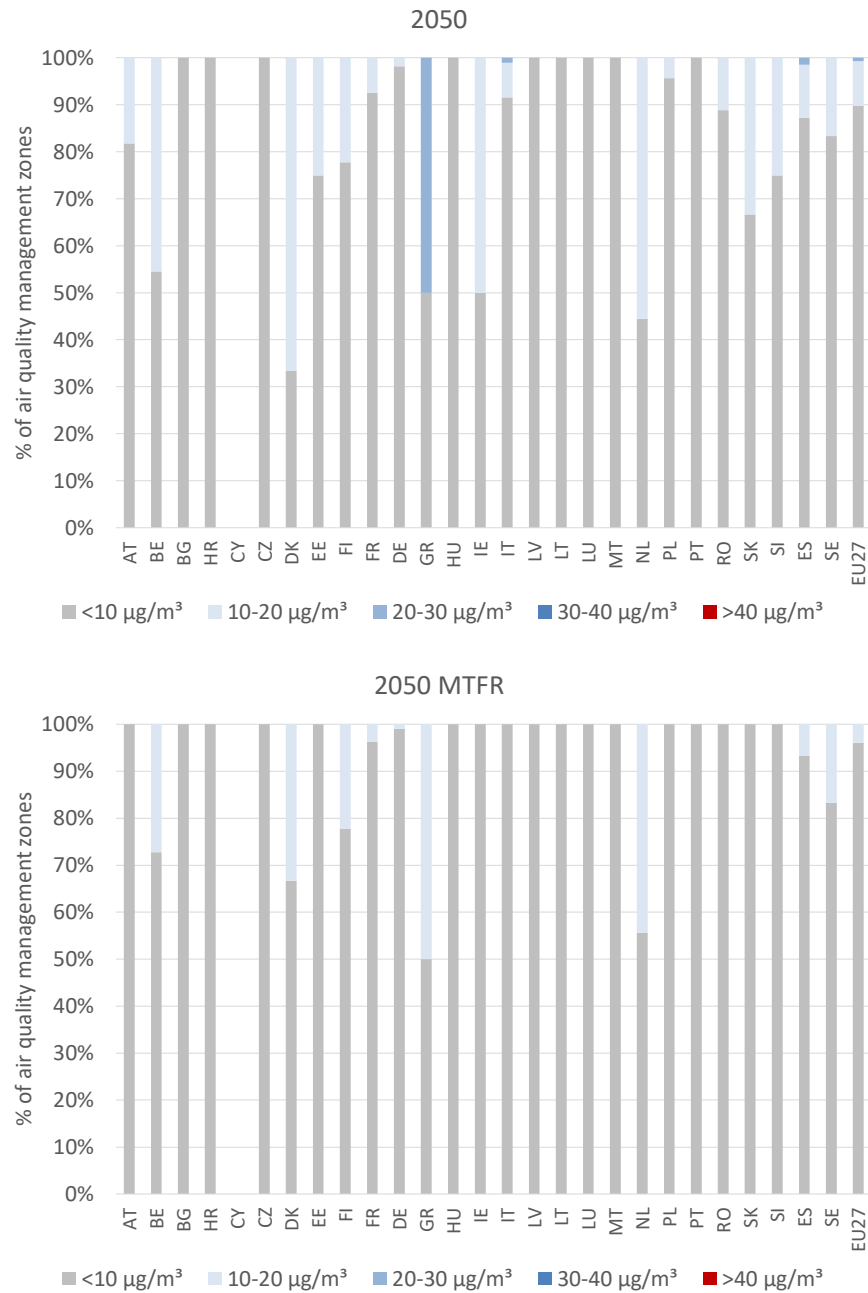


Figure 3.8: Distribution of maximum background concentrations of NO₂ in the air quality management zones, 2050. Upper panel: CAO2 baseline scenario, lower panel: CAO2 baseline with maximum technical air pollution controls. Cyprus is outside the geographical model domain.

3.3 Health impact indicators

The GAINS model estimates health impact indicators along the health impact assessment methodology that has been recommended for Europe by the World Health Organization (WHO) in its HRAPIE (Health risks from air pollution in Europe) project (WHO Regional Office for Europe, 2013).

3.3.1 Indicators for exposure to PM_{2.5}

With a specific focus on the epidemiological evidence gained in Europe and comparable regions, the HRAPIE (Health risks from air pollution in Europe) project of the WHO recommended to estimate health risks from long-term exposure to PM_{2.5}:

- for all-cause mortality,
- considering only population above 30 years age,
- employing long-term (annual mean) PM_{2.5} levels that are typical for the exposure of a given population (e.g., PM_{2.5} levels measured at urban background stations, but not at hot spots),
- apply a linear exposure-response function to all PM_{2.5} caused by anthropogenic sources, and
- exclude exposure from natural sources of PM_{2.5} (e.g., soil dust) in the assessment.

Depending on the focus of an assessment, other studies apply different methods that can deliver different results. For instance, the Global Burden of Disease (GBD) project, with its global focus, estimates specific health outcomes (e.g., lung cancer, acute lower respiratory infections, cardiovascular mortality, cardiopulmonary mortality, etc.), which, however, do not capture all diseases for which exposure to ambient air pollution is considered as a risk factor. To facilitate plausible results for rather high exposure levels (above 100 µg/m³) that are often found in developing countries, GBD employs non-linear 'Integrated Exposure Response' functions to various health outcomes that deliver lower responses at high exposure levels. However, the HRAPIE project found that for concentrations in the range where epidemiological evidence exists (i.e., up to ~30 µg/m³ PM_{2.5}) and which are typical for Europe, linear associations between exposure and response deliver a better statistical fit. Also, the GBD assessments apply a general lower threshold of 3-7 µg/m³ to total PM_{2.5} including the fraction originating from natural sources. As a consequence, GBD assessments suggest lower numbers than those computed with the WHO/HRAPIE method used in this report.

In contrast, the figures calculated by GAINS are somewhat lower than the analyses of the European Environment Agency (EEA). This is mainly due to the way in which exposure to PM_{2.5} from natural sources is taken into account (this cannot be readily quantified from the available air quality monitoring data unless the chemical composition of PM_{2.5} is analysed as well) and other methodological differences, including the geographical scale of the analysis.

Against this background, the GAINS analysis computes the following indicators for health impacts from PM_{2.5}:

- Mean population exposure to PM_{2.5} in each Member State, computed at a 7 km*7 km spatial resolution.
- Annual cases of premature deaths attributable to exposure to PM_{2.5}. To preserve consistency with the calculations performed for the NEC Directive analyses which used constant 2010 population in order to disentangle the impacts of population development and environmental policies, these calculations are performed also for the future with the 2010 population figures. Thereby, future demographic changes do not affect the distribution of mitigation efforts among Member States. However, the use of constant 2010 figures introduces inaccuracies in the assessment of future

health benefits, which scale with the change in total population. Thus, the accompanying assessment of the economic and health benefits from the CAO2 policy scenarios employs future population data that are projected for the different Member States.

- Alternatively, GAINS also provides the loss in statistical life expectancy that is attributable to a given exposure to PM_{2.5}. This refers to the population older than 30 years in 2010 considering their remaining lifetime over the next 100 years. While this has been found by the HRAPIE project as the most meaningful metric, experience showed that it was difficult to communicate and hard to understand for the general public and decision makers.
- GAINS also provides the Years of Life Lost (YOLLs) attributable to the exposure to a given level of PM_{2.5}, in terms of the cumulative number of life years lost that occur over the remaining life time of the population over 30 years living in 2010. Again, this excludes population dynamics, and is notably different from the annual YOLL estimates (in a particular year), which are most relevant for an economic assessment of the (annual) benefits of pollution control measures. Estimates of the annual YOLLs that take into account the population dynamics in the various Member States are provided in the accompanying benefits assessment report.

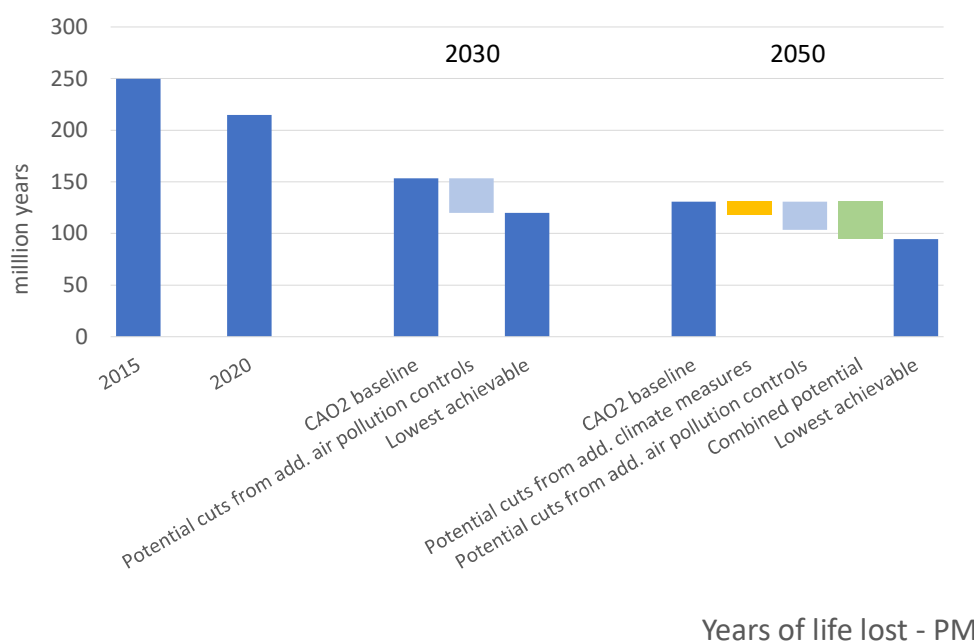
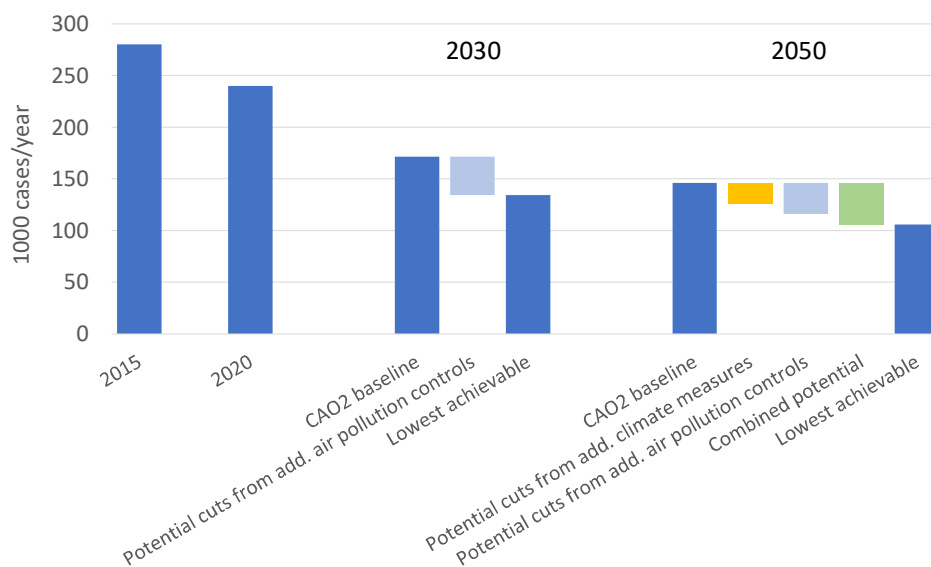
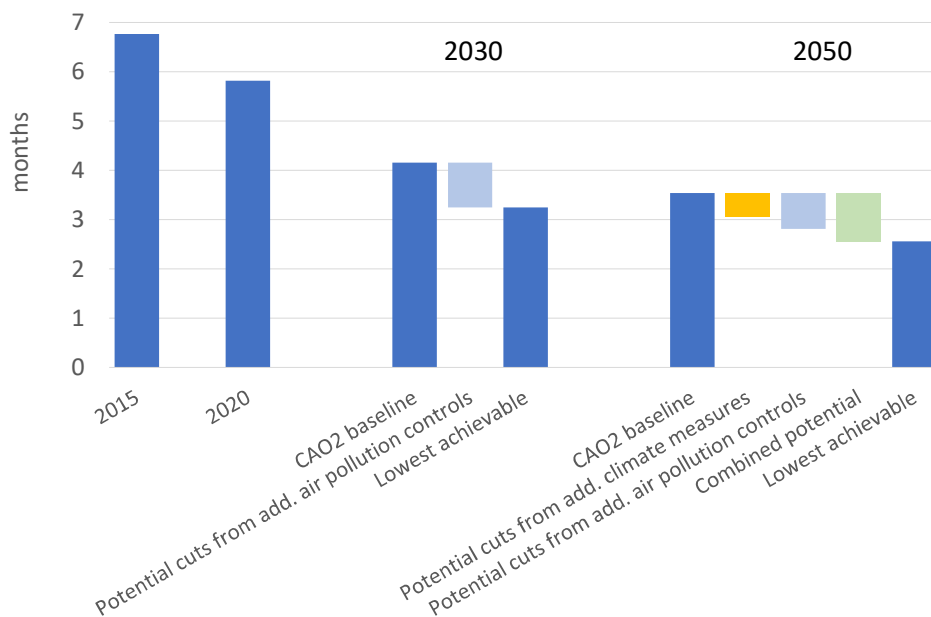


Figure 3.9: Years of life lost (YOLLs) attributable to the exposure to PM_{2.5} in the EU-27



Premature deaths - PM

Figure 3.10: Cases of premature deaths attributable to the exposure to PM_{2.5} in the EU-27



Loss in life expectancy - PM

Figure 3.11: Loss in statistical life expectancy from the exposure to PM_{2.5} in the EU-27

Table 3.8: PM_{2.5}-related health impact indicators for the policy scenarios for the EU-27

	2005	2010	2015	2020	2025	2030	2035	2040	2045	2050
Premature deaths attributable to PM_{2.5} (annual cases, using constant 2010 population data)										
Baseline scenarios										
CAO2 baseline	382,870	327,195	280,050	239,852	208,859	171,472	155,897	151,554	148,554	146,056
NAPCP					204,234	164,465	149,856	145,900	143,019	140,630
Climate policy scenarios										
1.5 TECH+WAM							145,815	141,770	139,978	137,160
1.5 LIFE+WAM							141,676	135,942	131,667	125,941
Air pollution control scenarios										
CAO2 baseline+MTFR						134,198				116,088
1.5 LIFE+MTFR										105,644
Years of life lost (YOLLs) attributable to PM_{2.5} (million years lost annually, using constant 2010 population data)										
Baseline scenarios										
CAO2 baseline	342.3	291.6	249.7	214.8	186.8	153.4	139.3	135.5	132.8	130.7
NAPCP					182.7	147.1	133.9	130.4	127.8	125.7
Climate policy scenarios										
1.5 TECH+WAM							130.3	126.7	125.2	122.7
1.5 LIFE+WAM							126.6	121.5	117.7	112.7
Air pollution control scenarios										
CAO2 baseline+MTFR						119.9				103.7
1.5 LIFE+MTFR										94.5

Table 3.9: PM_{2.5}-related health impact indicators for the policy scenarios for the EU-27

	2005	2010	2015	2020	2025	2030	2035	2040	2045	2050
Loss in statistical life expectancy attributable to PM_{2.5} (months)										
Baseline scenarios										
CAO2 baseline	9.3	7.9	6.8	5.8	5.1	4.2	3.8	3.7	3.6	3.5
NAPCP					5.0	4.0	3.6	3.5	3.5	3.4
Climate policy scenarios										
1.5 TECH+WAM							3.5	3.4	3.4	3.3
1.5 LIFE+WAM							3.4	3.3	3.2	3.1
Air pollution control scenarios										
CAO2 baseline+MTFR						3.3				2.8
1.5 LIFE+MTFR										2.6
Population-weighted PM_{2.5} concentrations (µg/m³)										
Baseline scenarios										
CAO2 baseline	15.2	12.9	11.0	9.5	8.3	6.8	6.2	6.0	5.9	5.8
NAPCP					8.1	6.5	5.9	5.8	5.7	5.6
Climate policy scenarios										
1.5 TECH+WAM							5.8	5.6	5.5	5.4
1.5 LIFE+WAM							5.6	5.4	5.2	5.0
Air pollution control scenarios										
CAO2 baseline+MTFR						5.3				4.6
1.5 LIFE+MTFR										4.2

3.3.2 Indicators for exposure to ground-level ozone

For ground-level ozone, GAINS provides the following indicators for health impacts that have been recommended by the WHO/HRAPIE project:

- The SOMO35 indicator is used as the central exposure metric recommended by the WHO HRAPIE project for assessing the health impacts attributable to the exposure to ground-level ozone in Europe. SOMO35 is calculated as the sum of the daily exceedances of maximum 8-hour ozone concentrations over a 35 ppb threshold, summed over the full year. SOMO35 values are computed with a spatial 0.5° longitude * 0.5° latitude resolution, and then multiplied with the population that is exposed in each of these grid cells.
- Annual cases of premature deaths attributable to the exposure to ground-level ozone, using constant 2010 population data.

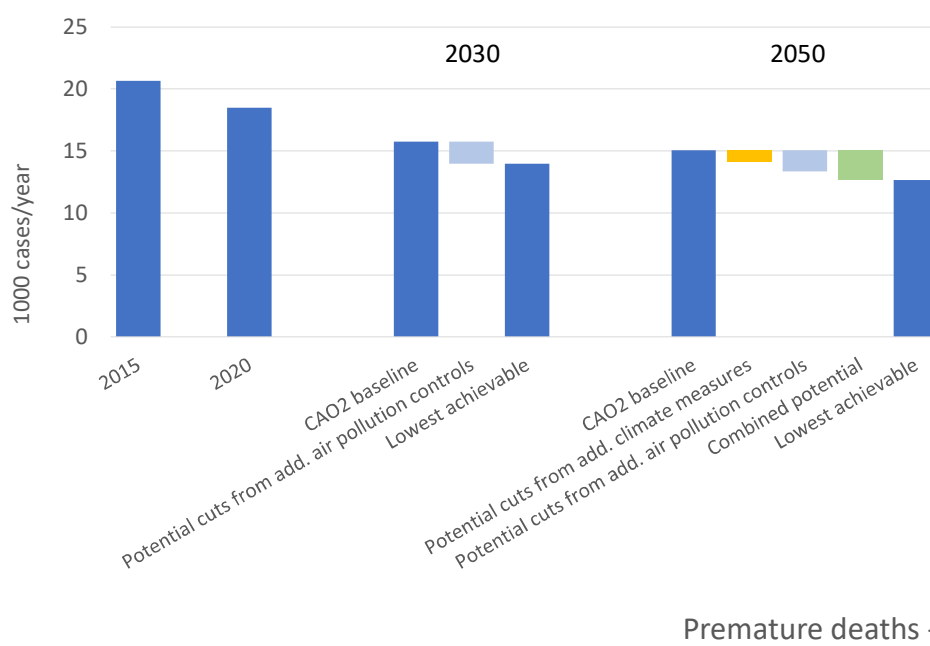


Figure 3.12: Cases of premature deaths attributable to the exposure to ground-level ozone in the EU-27

Table 3.10: Ground-level ozone-related health impact indicators for the policy scenarios for the EU-27

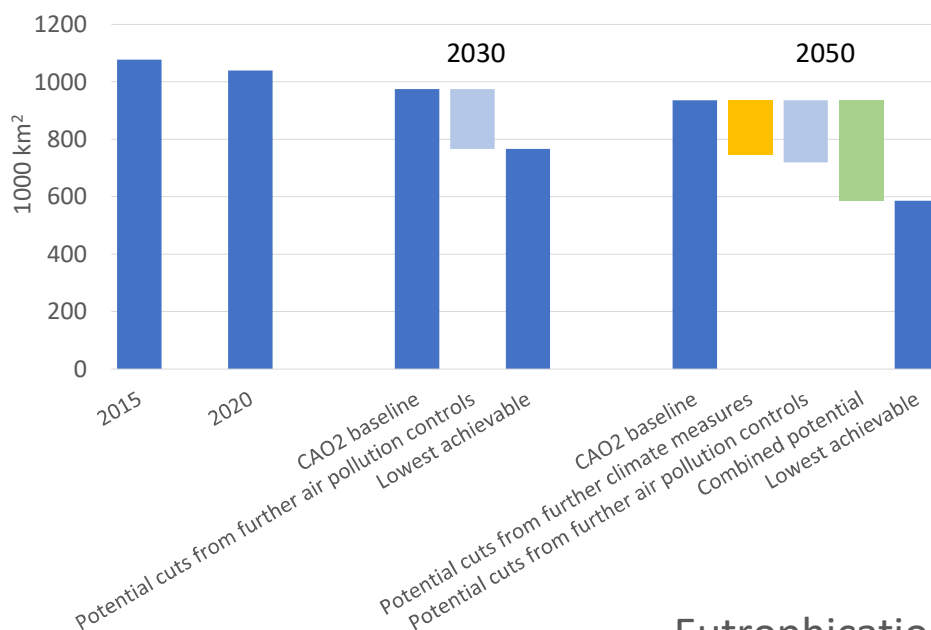
	2005	2010	2015	2020	2025	2030	2035	2040	2045	2050
Premature deaths attributable to ground-level ozone (annual cases, using constant 2010 population data)										
Baseline scenarios										
CAO2 baseline	26,123	22,997	20,646	18,494	16,967	15,753	15,221	15,129	15,041	15,040
NAPCP					16,942	15,723	15,202	15,120	15,034	15,035
Climate policy scenarios										
1.5 TECH+WAM							14,917	14,691	14,543	14,441
1.5 LIFE+WAM							14,874	14,612	14,337	14,113
Air pollution control scenarios										
CAO2 baseline+MTFR						13,960				13,356
1.5 LIFE+MTFR										12,653
Population-weighted SOMO35 (ppb.days)										
Baseline scenarios										
CAO2 baseline	3,948	3,476	3,122	2,802	2,570	2,388	2,310	2,298	2,287	2,287
NAPCP					2,566	2,383	2,307	2,297	2,286	2,286
Climate policy scenarios										
1.5 TECH+WAM							2,265	2,234	2,214	2,199
1.5 LIFE+WAM							2,259	2,222	2,183	2,150
Air pollution control scenarios										
CAO2 baseline+MTFR						2,121				2,038
1.5 LIFE+MTFR										1,933

3.4 Ecosystems impact indicators

The evolution of the standard indicators that have been used in the analyses for the NEC Directive and the Second Clean Air Outlook for ecosystems impacts for the various policy scenarios are shown in Table 3.11 to Table 3.14, and the potential changes from further climate and air policies and measures in Figure 3.13 to Figure 3.16. The assessment is fully compatible with the first Clean Air Outlook, as it employs the same critical loads data, i.e., those approved by the Executive Body of the LRTAP Convention in fall 2017 for use in policy assessments. Critical loads (CLs) are steady-state deposition thresholds and are derived for the impact of nitrogen (N) in terms of eutrophication and the combined impact of sulphur (S) and N deposition on the acidity status of an ecosystem; and their value depends on the properties of the that ecosystem. A CL is the maximum deposition rate of N (and S) compounds that do not lead to negative impacts in the long run. If deposition(s) are higher than the CL of an ecosystem, the CL is said to be exceeded (see Hettelingh et al., 2017 for more details).

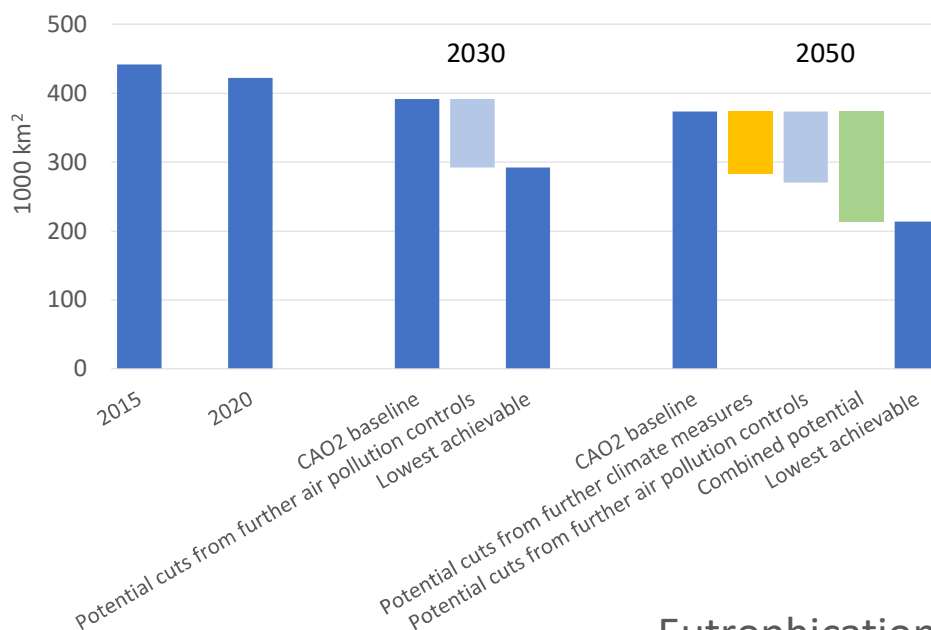
In particular, the following indicators are presented here:

- The total area of Natura2000 ecosystem protection zones in the EU-27 where nitrogen deposition exceeds the critical loads for eutrophication (in km² and as percentage of the total Natura2000 area). Only terrestrial ecosystems are included in the analysis.
- The total ecosystem area in the EU-27 where nitrogen deposition exceeds the critical loads for eutrophication (in km² and as percentage of the total Natura2000 area).
- Forest area in the EU-27 where acid deposition exceeds the critical loads for acidification (in km² and as percentage of the total forest area in the EU-27).
- The mean excess flux of ozone for deciduous trees in the EU-27. This is computed based on the POD1 metric for the phytotoxic ozone dose above a threshold of 1 nmol/m²/s that has been developed under the Convention on Long-range Transboundary Air Pollution. The mean excess flux is expressed in mmol/m².



Eutrophication - all ecosystems

Figure 3.13: Ecosystems area (1000 km²) where nitrogen deposition exceeds the critical loads for eutrophication, EU-27



Eutrophication - Natura2000 areas

Figure 3.14: Natura2000 nature protection areas (1000 km²) where nitrogen deposition exceeds the critical loads for eutrophication, EU-27

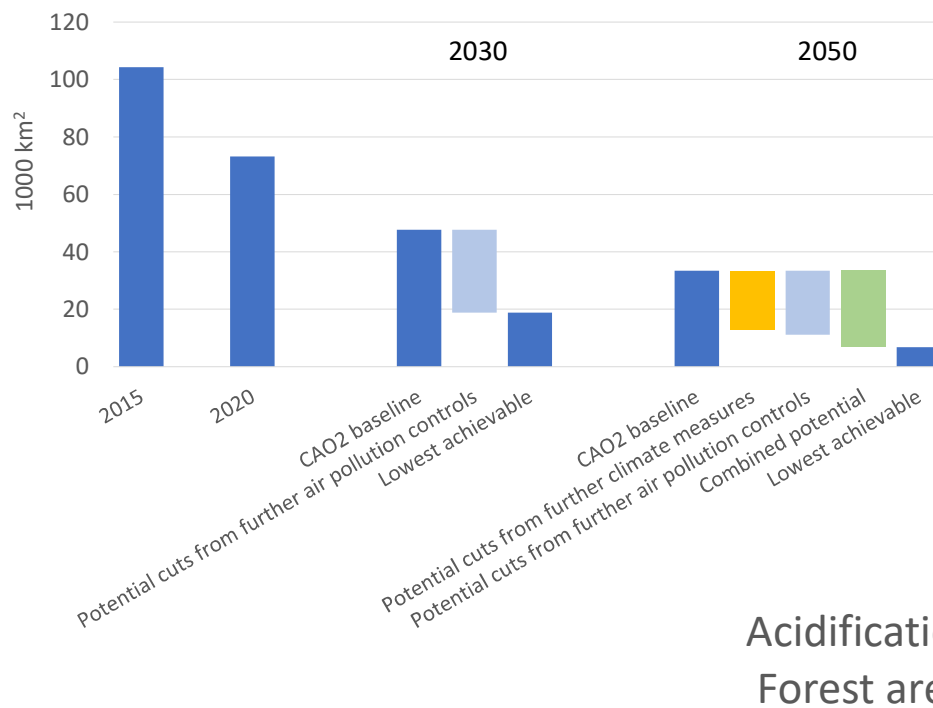


Figure 3.15: Forest area (1000 km²) where acid deposition exceeds the critical loads for acidification, EU-27

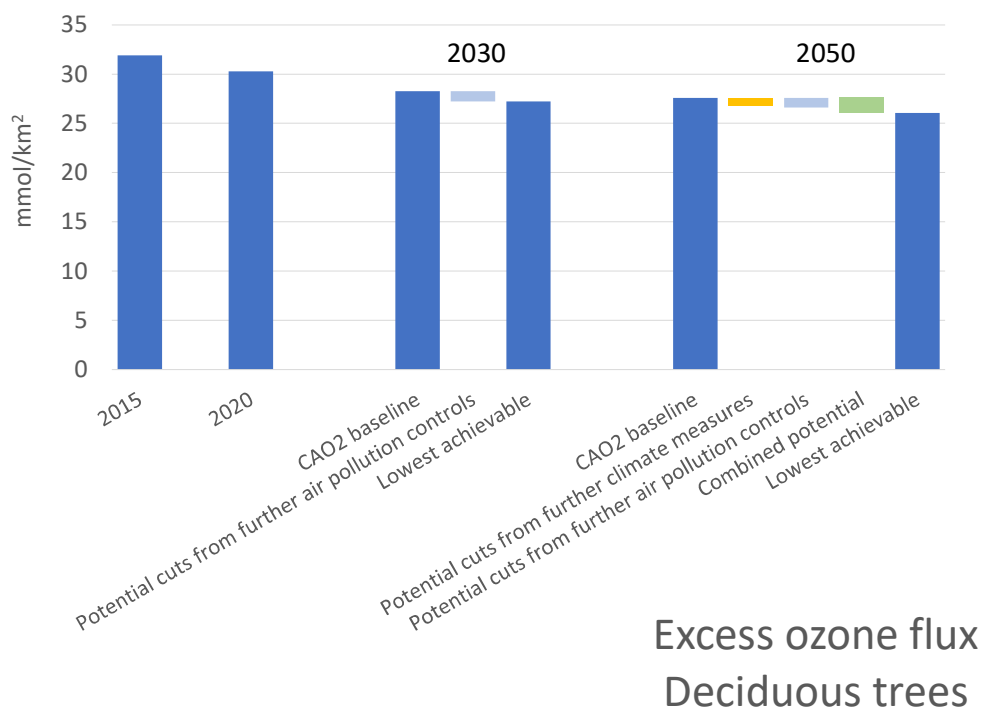


Figure 3.16: Excess ozone flux (mmol/km²) for deciduous trees, EU-27

Table 3.11: Natura2000 nature protection areas with nitrogen deposition exceeding the critical loads for eutrophication

	2005	2010	2015	2020	2025	2030	2035	2040	2045	2050
Natura2000 nature protection areas with nitrogen deposition exceeding the critical loads for eutrophication (1000 km²)										
Baseline scenarios										
CAO2 baseline	469.8	452.8	441.4	420.8	403.2	388.1	378.0	373.3	371.0	370.8
NAPCP					388.1	359.2	347.7	344.3	340.9	340.2
Climate policy scenarios										
1.5 TECH+WAM							343.2	337.6	335.4	334.1
1.5 LIFE+WAM							316.0	303.5	292.6	281.6
Air pollution control scenarios										
CAO2 baseline+MTFR						286.3				265.3
1.5 LIFE+MTFR										211.9
Natura2000 nature protection areas with nitrogen deposition exceeding the critical loads for eutrophication (% of total Natura2000 areas)										
Baseline scenarios										
CAO2 baseline	75.6%	72.9%	71.0%	67.7%	64.9%	62.5%	60.8%	60.1%	59.7%	59.7%
NAPCP					62.5%	57.8%	56.0%	55.4%	54.9%	54.8%
Climate policy scenarios										
1.5 TECH+WAM							55.2%	54.3%	54.0%	53.8%
1.5 LIFE+WAM							50.9%	48.8%	47.1%	45.3%
Air pollution control scenarios										
CAO2 baseline+MTFR						46.1%				42.7%
1.5 LIFE+MTFR										34.1%

Table 3.12: Ecosystems area with nitrogen deposition exceeding the critical loads for eutrophication, all ecosystems

	2005	2010	2015	2020	2025	2030	2035	2040	2045	2050
Ecosystems area with nitrogen deposition exceeding the critical loads for eutrophication, all ecosystems (1000 km²)										
Baseline scenarios										
CAO2 baseline	1132.7	1098.6	1076.6	1036.6	1001.3	968.2	946.5	936.3	930.4	929.5
NAPCP					971.2	910.6	884.9	877.5	870.0	868.5
Climate policy scenarios										
1.5 TECH+WAM							874.9	863.5	858.2	854.9
1.5 LIFE+WAM							822.3	792.3	768.1	744.0
Air pollution control scenarios										
CAO2 baseline+MTFR						754.3				708.4
1.5 LIFE+MTFR										581.7
Ecosystems area with nitrogen deposition exceeding the critical loads for eutrophication, all ecosystems (% of all ecosystems)										
Baseline scenarios										
CAO2 baseline	83.4%	80.9%	79.3%	76.3%	73.7%	71.3%	69.7%	68.9%	68.5%	68.4%
NAPCP					71.5%	67.0%	65.1%	64.6%	64.0%	63.9%
Climate policy scenarios										
1.5 TECH+WAM							64.4%	63.6%	63.2%	62.9%
1.5 LIFE+WAM							60.5%	58.3%	56.5%	54.8%
Air pollution control scenarios										
CAO2 baseline+MTFR						55.5%				52.2%
1.5 LIFE+MTFR										42.8%

Table 3.13: Forest area with acid deposition exceeding the critical loads for acidification

	2005	2010	2015	2020	2025	2030	2035	2040	2045	2050
Forest area with acid deposition exceeding the critical loads for acidification (1000 km²)										
Baseline scenarios										
CAO2 baseline	169.6	130.6	104.0	72.2	58.9	46.7	39.6	36.0	33.7	32.6
NAPCP					52.4	35.2	29.2	27.0	25.6	24.7
Climate policy scenarios										
1.5 TECH+WAM							26.1	24.0	22.6	21.4
1.5 LIFE+WAM							21.5	17.1	15.4	12.8
Air pollution control scenarios										
CAO2 baseline+MTFR						18.2				10.8
1.5 LIFE+MTFR										6.8
Forest area with acid deposition exceeding the critical loads for acidification (% of total forest area)										
Baseline scenarios										
CAO2 baseline	19.9%	15.3%	12.2%	8.5%	6.9%	5.5%	4.6%	4.2%	3.9%	3.8%
NAPCP					6.1%	4.1%	3.4%	3.2%	3.0%	2.9%
Climate policy scenarios										
1.5 TECH+WAM							3.1%	2.8%	2.7%	2.5%
1.5 LIFE+WAM							2.5%	2.0%	1.8%	1.5%
Air pollution control scenarios										
CAO2 baseline+MTFR						2.1%				1.3%
1.5 LIFE+MTFR										0.8%

Table 3.14: Excess ozone flux (phytotoxic ozone dose) above a threshold of 1 nmol/m²/s for deciduous trees (mmol/m²), EU-27

	2005	2010	2015	2020	2025	2030	2035	2040	2045	2050
Ozone fluxes (mmol/m²) for deciduous trees above a threshold of 1 nmol/m²/s										
Baseline scenarios										
CAO2 baseline	35.5	33.5	31.9	30.2	29.0	28.1	27.6	27.5	27.4	27.4
NAPCP					29.0	28.1	27.6	27.5	27.4	27.4
Climate policy scenarios										
1.5 TECH+WAM							27.4	27.2	27.1	27.0
1.5 LIFE+WAM							27.4	27.2	26.9	26.8
Air pollution control scenarios										
CAO2 baseline+MTFR						27.0				26.4
1.5 LIFE+MTFR										26.0

3.5 Economic valuation and economic analysis

3.5.1 Valuation methods

The methods for the assessment of health impacts and associated values are reviewed in Annexes 1.3 and 1.4, and conclusions have been integrated to the ALPHA-Riskpoll (ARP) model (Holland et al, 2013) which has been extensively used alongside GAINS in earlier cost-benefit analysis (CBA) of EU policies on air quality (e.g., Amann et al, 2017). The Annexes review practice in health impact assessment and CBA by international organisations including the EU, OECD, WHO and USEPA, and by analysts in various European countries.

The recommendations of the HRAPIE (Health Response to Air Pollutants In Europe) study led by WHO-Europe in 2013 (WHO, 2013) remain the most recent comprehensive review of air pollution epidemiology in Europe. Since HRAPIE was completed, however, there has been a substantial expansion of literature in the field and it became appropriate to consider updating the earlier methods with respect to effects and response functions previously included in the analysis, and with respect to the addition of further impacts. Annex 1.3 concludes that there are good grounds for considering that the HRAPIE recommendations underestimate the health impacts of air pollution. Several recent studies reviewed in Annex 1.3 consider additional impacts based on recent epidemiological literature (for example on coronary heart disease, stroke, asthma and diabetes) that would add significantly to the benefits from cleaner air quantified under HRAPIE. The WHO Systematic Review for PM_{2.5} (Chen and Hoek, 2020), and Pope et al (2020) also indicate that the HRAPIE function for mortality effects of long-term exposure to PM_{2.5} is conservative. This is particularly important as the effect of PM_{2.5} on mortality is dominant across CBA studies in this field. There is, however, no consensus across the studies reviewed on which additional effects should be included or, for mortality, which precise response function should be adopted. In the context of the present analysis, the following positions are adopted:

- The HRAPIE function set remains in use for the core estimates. This is acknowledged as a conservative position, biased to underestimation of health impacts. This position also enables comparison with earlier reports.
- Supplementary analysis is used to provide an indication of possible levels of underestimation of impacts by using the HRAPIE functions. The supplementary analysis applied here includes:
 - An increased estimate of PM_{2.5} related mortality, using the relative risk of 1.08 per 10 µg/m³ overall estimate from Chen and Hoek (2020), compared to 1.06 per 10 µg/m³ from HRAPIE. This is applied in the supplementary analysis rather than core because of variability in estimates from the Chen and Hoek (2020) and Pope et al (2020) studies. The overall estimate from Chen and Hoek is preferred to their estimate based on European studies only, given the much larger number of studies included in the former.
 - Additional response functions for stroke and cardiovascular disease via incidence of non-fatal myocardial infarction linked to PM_{2.5} exposure.

Inclusion of additional response functions for childhood asthma and diabetes was considered, though these effects were only included in one source (Ricardo, 2019). They have not been quantified here. The list of impacts considered in the health impact assessment is shown in Table 3.16.

With respect to valuation, the development of recommendations for updating the unit values is not straightforward, given a lack of consistency in the studies reviewed and the diversity of health metrics (values per case of new incidence, per prevalent case, per day, etc.) covering different effects. The review

provided in Annex 1.4 recommended the following values (values used in the analysis supporting the First Clean Air Outlook are also presented for ease of comparison).

Table 3.15. Recommended values for use in the Second Clean Air Outlook. Units: EUR, price year 2005, with prices for 2015 for consistency with the latest cost estimates from GAINS shown in brackets.

Effect	Figure used in analysis for 1 st CAO	Updated figures 2005, (2015), recommended for and used in 2 nd CAO	Main source(s)
Effects included by HRAPIE			
Mortality – value of statistical life (VSL)	€1.09 million (median) and €2.22 million (mean)	€3.06 million (€3.6 million)	Based on OECD (2012)
Mortality – value of a life year (VOLY)	€57,700 (median) and €138,700 (mean)	€79,500 (€94,660)	Previous median estimate increased in proportion to the increase in mean VSL to reflect OECD (2012)
Infant Mortality (per death)	€1.6 million (median) and €3.3 million (mean)	€4.53 million (€5.5 million)	Lower bound of OECD (2012) (factor 1.5 higher than average for adults)
Chronic Bronchitis in adults (per case)	€53,600	€53,600 (63,800)	Maca (2011), Holland (2014) with concerns over severity of air pollution related bronchitis
Bronchitis in children (per event)	€588	€301 (€358)	Hunt et al (2016)
Respiratory Hospital Admissions (per case)	€2,220	€4,000 (€4,800)	Broadly mid-range from estimates and similar to DCE (2018)
Cardiac Hospital Admissions (per case)	€2,220	€5,000 (€5,900)	Broadly mid-range from estimates and similar to DCE (2018)
Restricted Activity Days (per day)	€92	€110 (€131)	Hunt et al (2016)
Minor restricted activity days (per day)	€42	€40 (€48)	Hunt et al (2016)
Work loss days (per day)	€130	€130 (€155)	Amann et al (2017)
Asthma symptoms, asthmatic children (per day)	€42	€42 (€50)	Holland (2014), USEPA (2011)
Additional effects included here			
Stroke (per case)	Not quantified	€394,000 (€469,000)	Average of Åstrom (pers.com.) and Ricardo (2019)
Non-fatal myocardial infarction (per case)	Not quantified	€47,000 (€56,000)	Average of Åstrom (pers.com.) and USEPA (2011)

Valuation data have also been updated from a price year of 2005 to 2015, for consistency with the GAINS model. This update increases the earlier values by 19%, accounting for inflation and growth in real GDP per capita across the EU, and applying an income elasticity of 0.8, in line with recommendations from OECD (2012). The inflator used in the GAINS model is lower, 16.1%, but for valuation of the impacts it is appropriate to consider not only inflation but also growth in earnings and income elasticity of willingness to pay.

Four sensitivity cases are used for the CBA. The effects included for each case are shown in Table 3.16. The cases vary as follows:

- In use of VOLY or VSL for mortality valuation
- Consideration of additional effects on stroke and non-fatal myocardial infarction (included for supplementary analysis only)
- Inclusion of a higher response function for effects of chronic exposure to fine particles on mortality (increased from a relative risk of 1.06 to 1.08 $\mu\text{g.m}^{-3}$ for supplementary analysis only).

Table 3.16. Impact-valuation combinations forming the four sensitivity cases

Effect	Poll	Core +VOLY	Core + VSL	Supplementary + VOLY	Supplementary + VSL
Acute Mortality (deaths)	O ₃	✓	✓	✓	✓
Respiratory hospital admissions	O ₃	✓	✓	✓	✓
Cardiovascular hospital admissions	O ₃	✓	✓	✓	✓
Minor Restricted Activity Days	O ₃	✓	✓	✓	✓
Chronic Mortality – Life years lost	PM _{2.5}	✓			
Chronic Mortality - Deaths	PM _{2.5}		✓		
Infant Mortality - Deaths	PM _{2.5}	✓	✓	✓	✓
Chronic Bronchitis in adults	PM _{2.5}	✓	✓	✓	✓
Bronchitis in children aged 6 to 12	PM _{2.5}	✓	✓	✓	✓
Respiratory Hospital Admissions	PM _{2.5}	✓	✓	✓	✓
Cardiac Hospital Admissions	PM _{2.5}	✓	✓	✓	✓
Restricted Activity Days	PM _{2.5}	✓	✓	✓	✓
Asthma symptom days in children	PM _{2.5}	✓	✓	✓	✓
Lost working days	PM _{2.5}	✓	✓	✓	✓
Chronic Mortality – Life years lost	PM _{2.5}			✓	
Chronic Mortality - Deaths	PM _{2.5}				✓
Stroke	PM _{2.5}			✓	✓
Non-fatal myocardial infarction	PM _{2.5}			✓	✓

Some health impacts are modelled by both GAINS and the ARP model, though results are not directly comparable. The optimisation processes in GAINS account for change in risk, characterised through reference to a static population, which is appropriate for the needs of cost-effectiveness analysis (CEA). Including population change in GAINS would mean that differences in risk would be harder to interpret, being partly due to differences in population and partly to differences in pollution. ARP, in contrast, varies future population according to projections from World Population Prospects⁹ to account for changes in population size and age profile in order to be able to compare costs and benefits consistently (bearing in mind that underlying demand projections for polluting services such as energy and transport account for population change). There is a further difference in the way that mortality linked to chronic exposure to PM_{2.5} is reported. GAINS estimates life years lost under steady state conditions over the lifetime of the population (again, a risk-based approach appropriate to CEA), whilst ARP quantifies life years lost from one year's exposure in order to generate benefit estimates that can be compared with annual pollution control costs.

⁹ <https://population.un.org/wpp/>

3.5.2 Health benefits

Results of analysis for health impacts and associated values by scenario are shown in Annex 5. The valuation of health impacts is dominated by impacts associated with PM_{2.5} exposure and mortality, which accounts for between 69% and 93% of aggregate value of health damage (the range reflecting whether mortality is valued using the value of a life year or value of statistical life, and whether supplementary morbidity effects are included). Economic benefits relative to the CAO2 baseline scenario, extracted from the tables in the Annex, are shown in Table 3.17.

Table 3.17. Additional health benefits relative to the CAO2 baseline scenario, by scenario and sensitivity group (Core VOLY, Core VSL, Supplementary VOLY, Supplementary VSL). Units M€/year, 2015 prices.

Core VOLY	2025	2030	2035	2040	2045	2050
NAPCP	5,574	8,280	6,952	6,392	6,080	5,780
1.5 TECH+WAM			11,544	11,032	9,397	9,457
1.5 LIFE+WAM			16,292	17,487	18,341	21,177
CAO2 baseline+MTFR		44,424				31,572
1.5 LIFE + MTFR						42,558
Core VSL	2025	2030	2035	2040	2045	2050
NAPCP	20,506	31,974	28,220	27,556	27,927	28,061
1.5 TECH+WAM			46,952	47,363	42,793	45,443
1.5 LIFE+WAM			66,373	75,482	84,208	102,695
CAO2 baseline+MTFR		169,474				152,788
1.5 LIFE + MTFR						205,988
Supplementary VOLY	2025	2030	2035	2040	2045	2050
NAPCP	7,851	11,821	10,023	9,306	8,927	8,550
1.5 TECH+WAM			16,608	15,986	13,704	13,888
1.5 LIFE+WAM			23,460	25,390	26,837	31,198
CAO2 baseline + MTFR		63,060				46,502
1.5 LIFE + MTFR						62,658
Supplementary VSL	2025	2030	2035	2040	2045	2050
NAPCP	27,318	42,713	37,753	36,903	37,417	37,606
1.5 TECH+WAM			62,776	63,361	57,256	60,820
1.5 LIFE+WAM			88,760	101,014	112,732	137,510
CAO2 baseline + MTFR		226,096				204,586
1.5 LIFE + MTFR						275,795

For NAPCP the benefits relative to the CAO2 baseline increase up to 2030 but decline thereafter as measures in the Baseline come through later on. A similar pattern is seen for some of the other scenarios also, though the timing is different. For scenario 1.5 LIFE+WAM, benefits additional to the baseline go through to 2050. The climate scenarios clearly have very significant benefits, reducing air pollution impacts on health, though comparison of the results for 1.5 LIFE+WAM and 1.5 LIFE+MTFR in 2050 demonstrate that there are potentially significant benefits of further action on air pollution beyond the co-benefits linked to climate policies.

3.5.3 Reduced damages to materials, crops, forests and ecosystems

Damage to building materials was extensively researched in the 1980s and 1990s. Although research in the area continues, it is less active now than previously. Damage values per unit emission for SO₂ and NO_x have been taken from the CASES study (CASES, 2008).

The analysis of pollution damage to crops considers all agricultural crop production in the EU, focused on the effects of ozone. Methods are taken from the ECLAIRE study (Holland et al, 2015), drawing particularly on evidence generated through the Working Group on Effects under the Convention on Long

Range Transboundary Air Pollution. The tool developed in ECLAIRE has been updated for this analysis to use agricultural production in 2018 for the baseline. This is held constant through the period of assessment, though it is acknowledged that production patterns across Europe may change in the period to 2050 as a consequence of climate change and other changes. No account is taken of impacts that may affect production of animals and associated products (milk, eggs, honey, wool). The countries most affected by crop damages are France, Italy and Spain, and the most affected crops are wheat, olives, maize, grapes, rapeseed and tomato.

The analysis of forest damage takes account of ozone impacts leading to reduced productivity of wood for sale and of reduced carbon sequestration. Methods are again taken from the ECLAIRE study (Holland et al, 2015). Changes in productivity are valued against market data. Changes in carbon sequestration are valued using recommendations from DG Mobility and Transport's Handbook on the External Cost of Transport (EC, 2019). These recommendations are to apply marginal carbon mitigation costs of €100/t CO₂ up to 2030 and €269/tonne thereafter (2016 prices). Low and high variants were applied in the quantification of forest damage, with the Low estimate based on the application of carbon mitigation costs of €100/t CO₂ throughout, and the High estimate applying carbon mitigation costs of €269/t CO₂ from 2030 onwards.

Economic analysis of ecosystem damage is also based on the ECLAIRE study. It assesses impacts related to terrestrial ecosystems only. ECLAIRE used three methods to estimate damage costs for ecosystems, a willingness to pay (WTP) approach based on the results of research by Christie et al (2006, 2011, 2012), a repair cost approach described by Ott et al (Ott et al, 2006) and a 'regulatory revealed preference' approach (developed in the ECLAIRE study). The focus of analysis is on exceedance of the critical load for nitrogen in Natura 2000 sites, with valuation applied to the area subject to critical loads exceedance. No account was taken of exceedance of the critical load for acidification, because the area concerned is far less than that affected by eutrophication and there is potential for double counting if results for both effects are combined. The WTP based approach is adopted here as the preferred option because it follows a willingness to pay approach to valuation consistent with that used for other impacts assessed in this report. Uncertainties in the methods used in ECLAIRE led to the use of a factor 3 variation between low and high estimates of damage.

Detailed results for materials, crops, forests and ecosystems are presented in Annex 6. Table 3.18 summarises the benefits relative to the CAO2 baseline scenario for each of the other scenarios considered.

Table 3.18. Benefits from reduced damage to materials, crops, forests and natural ecosystems relative to CAO2 baseline for the EU27. Units: €million/year (2015 prices), LOW estimate.

	2025	2030	2035	2040	2045	2050
NAPCP	166	307	314	292	302	305
1.5 TECH + WAM			697	797	827	938
1.5 LIFE + WAM			977	1,155	1,302	1,546
CAO2 baseline + MTFR		2,305				2,248
1.5 LIFE + MTFR						3,312

Table 3.19. Benefits from reduced damage to materials, crops, forests and natural ecosystems relative to CAO2 baseline for the EU27. Units: €million/year (2015 prices), HIGH estimate.

	2025	2030	2035	2040	2045	2050
NAPCP	452	857	909	856	884	899
1.5 TECH + WAM			1,552	1,730	1,763	1,949
1.5 LIFE + WAM			2,382	2,788	3,183	3,766
CAO2 baseline + MTFR		4,236				5,056
1.5 LIFE + MTFR						7,515

3.5.4 Comparison of costs and benefits for health and non-health effects at the EU27 level

The cost of air pollution control measures generated by the GAINS model are shown in Table 3.20 and Table 3.21. Climate mitigation actions taken under Scenarios 1.5 TECH+WAM, 1.5 LIFE+WAM and 1.5 LIFE+MTFR lead to cost savings when considered against the costs of air pollution controls alone, by avoiding the need to undertake some level of air pollution abatement. Applying all measures in the GAINS model (the CAO2 baseline+MTFR scenario) leads to a large increase in abatement costs. However, these additional costs of MTFR are outweighed by the reduction in air pollution control costs under Scenario 1.5 LIFE+WAM (as shown by the negative result for Scenario 1.5 LIFE+MTFR in Table 3.21), though the saving is much reduced compared to 1.5 LIFE+WAM.

Table 3.20. Total air pollution costs for the EU27, by scenario. Units: €/M/year.

	2005	2010	2015	2020	2025	2030	2035	2040	2045	2050
CAO2 baseline	46,647	58,012	70,308	79,677	83,289	81,081	73,270	68,561	64,888	62,386
NAPCP				79,751	84,061	82,430	74,495	69,729	66,031	63,510
1.5 TECH + WAM							64,247	52,321	43,617	37,664
1.5 LIFE + WAM							63,891	51,823	42,969	37,314
CAO2Baseline+MTFR						106,710				85,993
1.5 LIFE + MTFR										57,589

Table 3.21. Incremental air pollution costs for the EU27 relative to CAO2 baseline, by scenario. Units: €/M/year.

	2025	2030	2035	2040	2045	2050
NAPCP	772	1,349	1,225	1,168	1,142	1,124
1.5 TECH + WAM			-9,379	-16,739	-21,920	-25,071
1.5 LIFE + WAM			-9,023	-16,241	-21,271	-24,722
CAO2 baseline + MTFR		25,629				23,607
1.5 LIFE + MTFR						-4,797

Total health and non-health benefits are shown in Table 3.22 for the four health valuation sensitivity cases shown in Table 3.16 (Core VOLY, Core VSL, Supplementary VOLY, Supplementary VSL), with benefits increasing through this series of sensitivity cases from Core VOLY (lowest benefit) to Supplementary VSL (highest benefit). The Core VOLY and Core VSL cases are combined with the Low estimate of non-health benefits, whilst the Supplementary VOLY and Supplementary VSL cases are combined with the high estimate.

Net benefits (benefits minus cost) are shown in Table 3.23. There is an increase in the values of benefits shown between Table 3.22 (total benefits) and Table 3.23 (net benefits) for scenarios 1.5 TECH+WAM, 1.5 LIFE+WAM and 1.5 LIFE+MTFR as a result of the cost savings (negative costs) for air pollution control for those scenarios shown in Table 3.21. Although climate policies are clearly beneficial for air quality, the increase in benefit when moving from scenario 1.5 LIFE+WAM to 1.5 LIFE+MTFR demonstrates that there is potential for significant further action to clear the air and improve health and the environment beyond those scenarios.

Table 3.22. Total benefits relative to CAO2 baseline for health and non-health effects for each scenario under the 4 sensitivity cases for health impact assessment and valuation. Units: €/year.

Core VOLY	2025	2030	2035	2040	2045	2050
NAPCP	5,739	8,587	7,267	6,684	6,382	6,086
1.5 TECH+WAM			12,241	11,829	10,224	10,395
1.5 LIFE+WAM			17,268	18,642	19,643	22,723
CAO2 baseline+MTFR		46,728				33,820
1.5 LIFE + MTFR						45,870
Core VSL	2025	2030	2035	2040	2045	2050
NAPCP	20,672	32,282	28,534	27,848	28,229	28,366
1.5 TECH+WAM			47,649	48,160	43,620	46,381
1.5 LIFE+WAM			67,349	76,637	85,510	104,241
CAO2 baseline+MTFR		171,779				155,036
1.5 LIFE + MTFR						209,300
Supplementary VOLY	2025	2030	2035	2040	2045	2050
NAPCP	8,303	12,678	10,932	10,162	9,811	9,448
1.5 TECH+WAM			18,160	17,717	15,467	15,837
1.5 LIFE+WAM			25,842	28,179	30,019	34,964
CAO2 baseline + MTFR		67,296				51,558
1.5 LIFE + MTFR						70,173
Supplementary VSL	2025	2030	2035	2040	2045	2050
NAPCP	27,770	43,570	38,662	37,759	38,301	38,505
1.5 TECH+WAM			64,327	65,091	59,018	62,769
1.5 LIFE+WAM			91,141	103,803	115,915	141,276
CAO2 baseline + MTFR		230,332				209,642
1.5 LIFE + MTFR						283,310

Table 3.23. Net benefits (benefits minus costs) relative to CAO2 baseline for each scenario under the four sensitivity cases for health impact assessment and valuation. Units: €/M/year.

Core VOLY	2025	2030	2035	2040	2045	2050
NAPCP	4,968	7,238	6,042	5,516	5,239	4,962
1.5 TECH+WAM			21,264	28,070	31,495	35,117
1.5 LIFE+WAM			26,647	35,381	41,562	47,795
CAO2 baseline+MTFR		21,099				10,212
1.5 LIFE + MTFR						50,667
Core VSL	2025	2030	2035	2040	2045	2050
NAPCP	19,900	30,932	27,309	26,680	27,086	27,243
1.5 TECH+WAM			56,672	64,401	64,891	71,103
1.5 LIFE+WAM			76,728	93,376	107,429	129,313
CAO2 baseline+MTFR		146,150				131,428
1.5 LIFE + MTFR						214,097
Supplementary VOLY	2025	2030	2035	2040	2045	2050
NAPCP	7,531	11,328	9,707	8,994	8,668	8,324
1.5 TECH+WAM			27,183	33,957	36,738	40,559
1.5 LIFE+WAM			35,221	44,917	51,939	60,036
CAO2 baseline + MTFR		41,667				27,951
1.5 LIFE + MTFR						74,970
Supplementary VSL	2025	2030	2035	2040	2045	2050
NAPCP	26,999	42,221	37,437	36,591	37,158	37,381
1.5 TECH+WAM			73,350	81,332	80,289	87,490
1.5 LIFE+WAM			100,520	120,541	137,834	166,347
CAO2 baseline + MTFR		204,703				186,035
1.5 LIFE + MTFR						288,107

These results show a significant surplus of benefit over cost for each scenario at the EU27 level. Under the NAPCP scenario, benefits start in the range of €4,968 to €26,999 million per year in 2025, increasing to between €7 and 42 billion per year in 2030 with a gradual decline thereafter as measures in the baseline gradually catch up. Net benefits of the climate policies are broadly between a factor 3.5 and 10 higher.

Results for scenarios 1.5 TECH+WAM and 1.5 LIFE+WAM demonstrate that climate policies will have major benefits in terms of reducing the air quality burden on health and other receptors. This is not surprising, having been shown by numerous previous studies. However, an understanding of the way that air and climate policies interact is critical to developing optimal policies addressing the two problems simultaneously.

Results for the NAPCP and the two MTFR scenarios demonstrate that there are significant benefits from additional measures to improve air quality beyond those that would arise from climate action only.

Care is needed in interpretation of the results for the MTFR scenarios as they could be interpreted as demonstrating a net benefit for all measures included for MTFR. In reality, some of the measures included in the cost curve will be cost-efficient whilst some others are extremely expensive per unit emission abated, and on their own may not generate a net benefit either across the EU27 or in all Member States. This is suggested by the decline in benefit-cost ratios when moving from NAPCP to Baseline+MTFR. More detailed analysis, involving a series of additional scenarios would be needed to test the costs and benefits of the increasingly less cost-effective measures towards the upper end of the cost curve.

Benefit-cost ratios are shown in Table 3.24. Cases involving cost savings (1.5 TECH+WAM, 1.5 LIFE+WAM and 1.5 LIFE+MTFR) are not presented as there is no additional 'cost' to consider in terms of air pollution controls.

Table 3.24. Benefit-cost ratios for each scenario under the 4 sensitivity cases for health impact assessment and valuation for scenarios NAPCP and CAO2 baseline+MTFR, relative to CAO2 baseline.

Core VOLY	2025	2030	2035	2040	2045	2050
NAPCP	7.44	6.36	5.93	5.72	5.59	5.42
CAO2 baseline+MTFR		1.82				1.43
Core VSL	2025	2030	2035	2040	2045	2050
NAPCP	26.79	23.92	23.29	23.85	24.71	25.24
CAO2 baseline+MTFR		6.70				6.57
Supplementary VOLY	2025	2030	2035	2040	2045	2050
NAPCP	10.76	9.40	8.92	8.70	8.59	8.41
CAO2 baseline + MTFR		2.63				2.18
Supplementary VSL	2025	2030	2035	2040	2045	2050
NAPCP	35.99	32.29	31.56	32.34	33.52	34.26
CAO2 baseline + MTFR		8.99				8.88

3.5.5 Macro-economic impacts

This section integrates the air pollution control costs and the corresponding benefits of clean air into the broader, economy-wide framework as captured by the JRC-GEM-E3 model. Accounting for interactions in the context of the full economic system is a valuable step in the analysis for several reasons. First, while abatement expenditures are a cost for one sector, these expenditures also represent increased demand for goods and services of the sectors that deliver the goods required in the abatement process. Second, both the costs and the increased demand for abatement goods may have impacts along the supply chain. These effects are included here, as the model builds on input-output tables of GTAP10 (Aguiar et al., 2019). Third, costs and benefits for an economic sector may be exacerbated or attenuated by international trade. Trade effects can amplify losses when a sector loses competitive advantage compared to foreign firms. Alternatively, trade can limit losses when a sector has the possibility to substitute expensive domestic inputs for imported ones. Fourth, general equilibrium feedbacks in relation to the labour market can influence the economic outcome. For instance, cleaner air leads to a reduction in sick leaves. When the implied increase in labour productivity translates into higher wages, households experience higher purchasing power and may raise overall consumption levels. Finally, the integrated economic perspective enables the analysis of economic effects at sector-specific level, providing more insight into how efforts and gains are shared between different actors in the economy.

To assess the macro-economic impacts of the pollution control policies that are analysed in the scenarios, country- and sector-specific abatement costs from the GAINS model are fed into the economy-wide JRC-GEM-E3 model (Figure 3.17). GEM-E3 is a Computable General Equilibrium (CGE) model, representing firms' production choices, households' consumption behaviour, international trade flows and government policy and budgetary constraints. Abatement costs are represented as additional intermediate inputs required in the production process. On the benefit side, two 'market' benefits, calculated in the Alpha-RiskPoll framework, serve as input into JRC-GEM-E3: crop yield impacts are translated into changes in total factor productivity of the crop sector, and changes in lost work days are represented as a corresponding change in labour productivity. The framework also includes a direct link from the Alpha-RiskPoll model to the economy-wide cost-benefit assessment, representing the non-market benefits that are valued outside of the economic modelling, as presented in the previous sections. This methodology follows the approach of the First Clean Air Outlook¹⁰. The framework is furthermore

¹⁰ https://ec.europa.eu/environment/air/clean_air/outlook.htm

comparable to the work done in the PESETA projects¹¹ on climate impacts, where a similar CGE model is used to synthesise impacts from changing climatic conditions as derived from bottom-up or biophysical models.

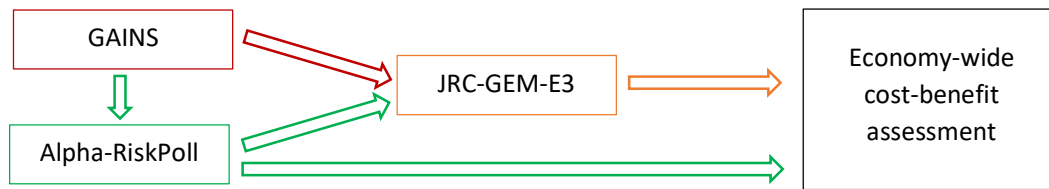


Figure 3.17: Methodological approach for the economy-wide cost-benefit assessment

Costs and benefits of five scenarios, after accounting for economy-wide interactions, are shown in Figure 3.18 (relative to the baseline). The gains outweigh costs in all scenarios, when all benefit categories are taken into account. Avoided mortality from reduced PM_{2.5} concentrations is the dominating benefit category. While the COVID-19 pandemic is a stark reminder of the importance of healthy lives, the assessment here illustrates the importance of considering health impacts also in other domains. The scenarios that reach climate neutrality in 2050 (1.5 TECH+WAM, 1.5 LIFE+WAM, and 1.5 LIFE+MTFR) highlight the co-benefits of climate policy in terms of reduced abatement costs for air pollution control measures. As the energy system would transition towards clean and renewables sources of energy, expenditures to control air pollution from the burning of fossil fuel would become unnecessary. Costs of climate policy and corresponding benefits in terms of avoided climate impact damages are not included in the analysis presented here.

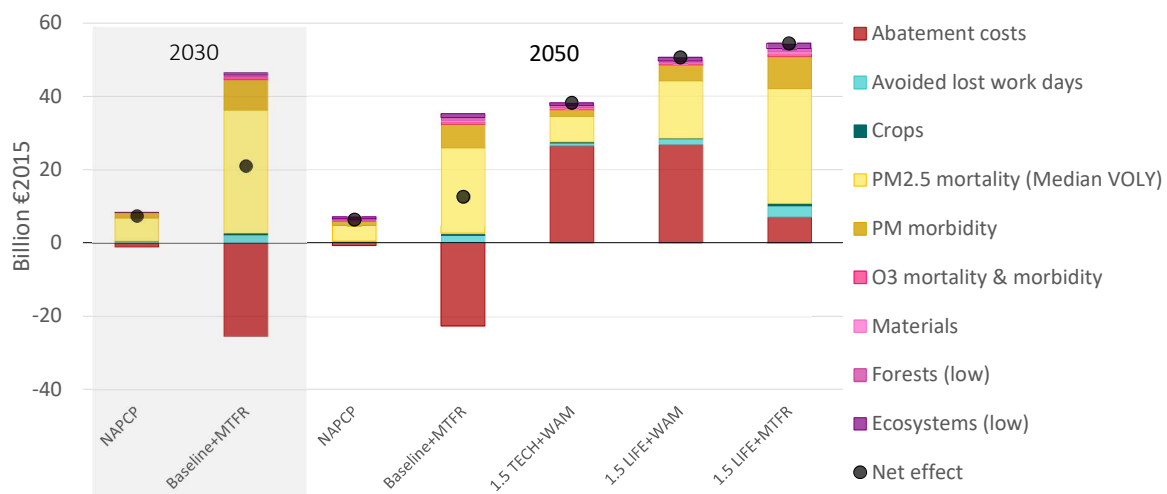


Figure 3.18: Cost-benefit assessment for the EU27, relative to the baseline. Source: GAINS, ALPHA-Riskpoll and JRC-GEM-E3 models.

¹¹ <https://ec.europa.eu/jrc/en/peseta-iv>

Air pollution control costs differ across sectors, in line with their emissions and the corresponding economic abatement potential (Figure 3.19). The scenarios with ambitious air pollution controls (Maximum Technically Feasible Reduction, MTFR) require additional investments for industry (manufacturing and construction) and livestock. Ambitious climate policies (1.5 TECH, 1.5 LIFE) lead to substantially reduced abatement expenditures compared to the baseline for the service sector and for households, as these sectors increase energy efficiency and end-use electrification.

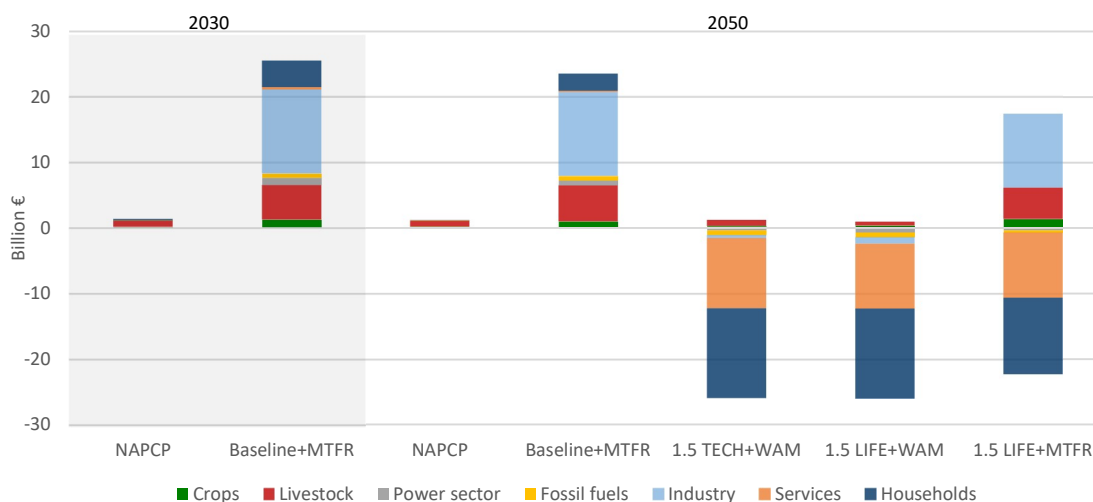


Figure 3.19: Air pollution control costs across sectors and scenarios, relative to the baseline, for the EU27. Source: GAINS model.

To provide a broader assessment of sector-specific impacts, the partial equilibrium costs have been fed into the economy-wide JRC-GEM-E3 model to account for interaction effects in the economic system. These include potential changes in international trade, impacts along the supply chain and via the labour market. Table 3.25 presents a view of the sector-specific economic implications of abatement costs before and after accounting for ‘market’ benefits (avoided lost work days and increased crop yields from Alpha-RiskPoll). An important consideration here is that abatement efforts require the purchasing of additional intermediate inputs in the production process, which represent additional demand mainly for industrial products required for abatement. For this reason, additional abatement expenditures in the industrial sector do not necessarily reduce aggregate output of industrial sectors, but may imply a shift of production at a finer level of disaggregation within the industrial sector. In the most optimistic case (1.5 LIFE+WAM), EU aggregate GDP would increase by 0.15% compared to the baseline in 2050. This GDP increase would offset around 10-50% of the costs of climate policy as assessed for the proposal for a Long-Term Strategy on climate action¹².

Increased crop yields due to cleaner air offset 54-95% of the output losses of the crop sector (including forestry) in the MTFR scenarios. In the 1.5°C scenarios, the value of enhanced crop yields covers 95-208% of the sector’s output losses, producing net gains overall. Under the NAPCP scenario, the share of crop sector output losses that is compensated by higher crop yields remains below 10%. The livestock sector consistently experiences output losses across scenarios, while the service sector output levels are higher than in the baseline for all cases considered. EU-wide production of the fossil fuel sectors, including

¹² In-depth analysis in support of COM(2018)773: A Clean Planet for all: A European long-term strategic vision for a prosperous, modern, competitive and climate neutral economy. High and low GDP costs taken from table 14.

extraction and refinery activities, decreases under the MTR scenarios, in line with additional abatement expenditures.

Recent empirical evidence (Dechezleprêtre et al., 2019) suggests that the macro-economic gains could be larger than the results presented above. Building on the work of Dechezleprêtre et al. (2019), two approaches to include this evidence in the analysis are identified: The first approach is based on the estimate that output per worker increases by 0.80% per 1 μgm^{-3} decrease in the concentration of fine particulate matter. This is consistent with recent findings for the impact of $\text{PM}_{2.5}$ on productivity in the manufacturing sector in the US (Chang et al., 2016; 0.8% per μgm^{-3}) and slightly higher than recent estimates for the service sector productivity in China (Kahn and Li, 2020; 0.35% per μgm^{-3}). We interpret this estimate as a response in labour productivity, while assuming a linear exposure-response function such that this point estimate applies across scenarios and regions, also for future years. The second approach builds on the estimate that GDP increases by 0.83% per 1 μgm^{-3} decrease in the concentration of $\text{PM}_{2.5}$. To translate this estimate into an input shock for the JRC-GEM-E3 model, we maintain consistency with the labour productivity shock of 0.80% per μgm^{-3} and complement this with an equal productivity shock for capital. As the econometric study of Dechezleprêtre et al. (2019) cannot pin down exactly the channels through which air pollution affects aggregate economic output, we have chosen here to use a shock to total factor productivity – labour and capital (including land) – as this has the advantage to reflect the empirical evidence of output per worker while replicating aggregate GDP impacts at the same time. Prior to running the scenarios based on this second approach, we calibrate the JRC-GEM-E3 model to the macro-economic estimates, using a small and uniform (across regions and capital and labour productivity) scaling factor to replicate the 0.83% of GDP per 1 μgm^{-3} exactly, after general equilibrium interactions. The corresponding productivity shocks are then combined with the changes in $\text{PM}_{2.5}$ concentrations. As the analysis is done at the Member State level, we do not study how region-specific changes in air pollution may coincide with the particular location of industries within a country. We furthermore abstract from potential productivity impact differences across sectors, although this could be included in future work if the empirical evidence is sufficiently strong.

Figure 3.17 illustrates that the magnitude of these estimates influences the cost-benefit assessment, which now yields positive GDP impacts across all scenarios. This is true even when only considering the labour productivity impact, which implicitly covers both the extensive (e.g., reduced sick leave and absence from work) and the intensive (on-the-job performance) margin of labour productivity. In the best case (1.5 LIFE+WAM), GDP increases by 1.3% compared to the baseline in 2050. In all climate policy scenarios, economic benefits linked to cleaner air thanks to climate and clean air policies offset all the air pollution control costs, plus a substantial share of climate policy costs. Previous estimates with the JRC-GEM-E3 model for GDP impacts of climate policy in the 1.5 TECH scenario range from -0.29% to -1.30% of GDP in 2050¹³, such that the gains presented here (1.5 TECH+WAM) would offset from 20% to well over 100% of these costs. Importantly, these are ‘market’ impacts in the sense that they translate into GDP numbers, without explicitly considering the non-market value of avoided mortality. Accounting for avoided mortality in a broader welfare metric, for instance through a willingness-to-pay approach, would further raise the benefits over the GDP estimate shown here.

Introducing recent empirical evidence to evaluate the benefits of cleaner air also has repercussions on the sectoral impact estimates. Crop sector outputs increase compared to the baseline in the majority of scenarios. Note that we use a uniform productivity effect across sectors, whereas the estimates of Dechezleprêtre et al. (2019) suggest that gains are likely higher in agriculture and construction. Impacts

¹³ In-depth analysis in support of COM(2018)773: A Clean Planet for all: A European long-term strategic vision for a prosperous, modern, competitive and climate neutral economy. High and low GDP costs taken from table 14.

on industry and service sector output are consistently positive across all scenarios. The results indicate that productivity gains from cleaner air outweigh policy costs for a large group of stakeholders. Future work may further explore the economic benefits from cleaner air, as a better understanding of these effects may further raise support for ambitious policy action

Table 3.25: Macro-economic and sector-specific impacts in 2030 and 2050, expressed as percentage changes compared to the baseline

Excluding including market benefits	NAPCP 2030	Baseline+MTFR 2030	NAPCP 2050	Baseline+MTFR 2050	1.5 TECH+WAM 2050	1.5 LIFE+WAM 2050	1.5 LIFE+MTFR 2050
GDP	-0.007 -0.004	-0.168 -0.150	-0.004 -0.002	-0.115 -0.102	0.134 0.139	0.136 0.145	0.036 0.055
Private consumption	-0.009 -0.005	-0.131 -0.110	-0.004 -0.001	-0.078 -0.063	0.145 0.15	0.143 0.154	0.082 0.102
Sector output							
Crops	-0.15 -0.13	-0.89 -0.38	-0.11 -0.11	-0.56 -0.07	-0.15 0.17	-0.18 0.02	-0.72 -0.01
Livestock	-0.45 -0.44	-2.75 -2.70	-0.41 -0.41	-2.65 -2.61	-0.41 -0.38	-0.26 -0.24	-2.32 -2.26
Power sector	0.06 0.06	0.54 0.55	0.05 0.05	0.49 0.50	0.07 0.07	0.06 0.07	0.51 0.52
Fossil fuels	0.00 0.00	-0.65 -0.64	0.00 0.00	-0.60 -0.60	0.17 0.17	0.19 0.19	-0.27 -0.27
Industry	0.01 0.01	-0.01 0.00	0.01 0.01	-0.03 -0.02	-0.17 -0.17	-0.16 -0.15	-0.20 -0.20
Services	0.00 0.00	0.00 0.02	0.00 0.00	0.01 0.02	0.08 0.08	0.07 0.08	0.09 0.10

Note: In each column, the numbers on the left-hand side indicate the impacts when the macro-economic benefits from clean air are not considered. The numbers on the right-hand side in each column include market benefits of cleaner air: avoided lost work days and crop yield benefits. Non-market benefits of clean air and costs of climate policy are not included.

Table 3.26: Macro-economic and sector-specific impacts based on recent empirical evidence, expressed as percentage changes compared to the baseline

Labour Total Factor Productivity	NAPCP 2030	Baseline+MTFR 2030	NAPCP 2050	Baseline+MTFR 2050	1.5 TECH+WAM 2050	1.5 LIFE+WAM 2050	1.5 LIFE+MTFR 2050
GDP	0.09 0.20	0.32 0.86	0.09 0.18	0.31 0.79	0.27 0.42	0.44 0.77	0.62 1.26
Private consumption	0.12 0.26	0.48 1.19	0.11 0.22	0.45 1.06	0.31 0.50	0.52 0.94	0.80 1.62
Sector output							
Crops	0.01 0.13	-0.11 0.49	0.03 0.14	0.16 0.71	0.07 0.24	0.31 0.69	0.24 0.98
Livestock	-0.32 -0.21	-2.14 -1.56	-0.30 -0.20	-2.09 -1.57	-0.23 -0.07	0.14 0.49	-1.56 -0.86
Power sector	0.15 0.28	1.01 1.72	0.13 0.24	0.92 1.54	0.20 0.38	0.35 0.76	1.08 1.91
Fossil fuels	0.08 0.20	-0.23 0.39	0.07 0.17	-0.25 0.28	0.28 0.44	0.43 0.79	0.21 0.92
Industry	0.14 0.26	0.59 1.22	0.12 0.23	0.51 1.06	0.00 0.17	0.22 0.60	0.52 1.26
Services	0.10 0.20	0.46 0.99	0.09 0.18	0.41 0.87	0.20 0.35	0.35 0.67	0.62 1.25

Note: In each column, the numbers on the left-hand side indicate the impacts when benefits from clean air for labour productivity are considered. The numbers on the right-hand side in each column include both labour and capital productivity gains. Non-market benefits of clean air and costs of climate policy are not included.

3.6 Results for the recent Mix55 climate policy scenario

In September 2020, after completion of the main scenario calculations for this report, the European Commission proposed a new 2030 Climate ambition, as part of the European Green Deal. The 2030 Climate Ambition communication proposes to cut greenhouse gas emissions by at least 55% by 2030, in order to set Europe on a responsible path to becoming climate neutral by 2050. This goal should be achieved through expansion of carbon pricing (ETS) and enhanced policies for renewables (a share of 38.4% in 2030) and energy efficiency (40% reduction in primary energy consumption).

The Commission proposal was supported by an Impact Assessment¹⁴ that presents *a central level of the climate target and interaction with energy policy* (the 'Mix55' scenario') to illustrate a concrete pathway towards the achievement of the targets.

This Section presents the clean air impacts of the Mix55 scenario, assuming also the additional policies and measures of the NAPCPs. However, due to more recent projections on economic growth and the different timing of climate targets, results are not directly comparable to the CAO2 scenarios.

Although numerical results are not directly comparable with the other scenarios discussed in this report, the Mix55 scenario results in a steeper decline of the health- and ecosystems impact indicators (Table 3.27).

Table 3.27: Changes of the impact indicators (relative to 2015) of the CAO2 baseline and Mix55 scenarios (EU-27)

	CAO2 baseline scenario		Mix55 scenario	
	2030	2050	2030	2050
Premature deaths				
due to PM2.5	-38.8%	-47.8%	-41.9%	-48.6%
due to ozone	-23.7%	-27.2%	-25.2%	-30.2%
Ecosystems impacts				
Eutrophication, all ecosystems	-10.1%	-13.7%	-10.9%	-22.4%
Eutrophication, Natura2000 areas	-12.1%	-16.0%	-13.4%	-26.8%
Acidification of forest soils	-55.1%	-68.7%	-57.4%	-79.3%
Ozone fluxes for forests	-11.9%	-14.1%	-12.9%	-15.4%

¹⁴ Commission Staff Working Document Impact Assessment Accompanying The Document Communication From The Commission To The European Parliament, The Council, The European Economic And Social Committee And The Committee Of The Regions: Stepping up Europe's 2030 climate ambition Investing in a climate-neutral future for the benefit of our people. SWD/2020/176 final

3.7 Cross-border effects of national policies and measures

To quantify the cross-border effects of national policies and measures, the GAINS model has been used to estimate for all Member States the impacts of the emission reductions that will occur within their own territory, and those that will emerge in the other European/EU countries.

3.7.1 Import-export budgets for PM_{2.5}

Quantifications of the contributions of emission (reductions) from each Member State on population-weighted mean exposure to PM_{2.5} can be presented in the form of “import-export” matrices for key scenarios. Thereby, the differences in the individual contributions between emission scenarios reflect the aggregated impact of the emission changes from all measures. The impacts of individual measures will vary to a certain extent, depending on their impact on the mixture of PM precursor emissions (primary PM_{2.5}), SO₂, NO_x, NH₃, NMVOC, as well as on the emission height of the source.

A PM_{2.5} dispersion budget for 2015 (i.e., how much each Member State contributed to population-weighted PM_{2.5} concentrations in a country) is shown in Table 3.28, while Table 3.30 provides data for the CAO2 baseline in 2030.

Table 3.28: Origin of population-weighted PM_{2.5} background concentrations in each Member State computed for 2015 (µg/m³)

		Source regions																										Other Europe	Intern. shipping	Natural + Africa	Total	
		AT	BE	BG	HR	CY	CZ	DK	EE	FI	FR	DE	GR	HU	IE	IT	LV	LT	LU	MT	NL	PL	PT	RO	SK	SI	ES					SE
Receptor regions	AT	3.31	0.03	0.01	0.12	0	0.52	0	0	0	0.17	1.24	0	0.27	0	0.5	0	0	0	0	0.03	0.65	0	0.07	0.11	0.24	0.01	0	0.62	0.09	1.73	9.72
	BE	0	4	0	0	0	0.1	0	0	0	1.96	1.86	0	0	0.05	0.03	0	0	0.04	0	0.91	0.26	0	0	0	0	0.04	0	1.02	0.48	1.57	12.32
	BG	0.01	0	5.62	0	0	0.05	0	0	0	0	0.1	0.18	0.08	0	0.06	0	0	0	0	0	0.21	0	0.83	0.03	0	0	0	2.54	0.16	3.52	13.39
	HR	0.27	0.01	0.03	3.25	0	0.3	0	0	0	0.08	0.4	0	0.52	0	0.86	0	0	0	0	0.01	0.55	0	0.2	0.13	0.35	0.02	0	3.58	0.25	2.27	13.08
	CY	0	0	0.04	0	0.97	0	0	0	0	0	0.01	0.14	0	0	0.03	0	0	0	0	0	0.01	0	0.03	0	0	0	0	3.74	0.94	7.02	12.93
	CZ	0.32	0.05	0.01	0.07	0	4.28	0	0	0	0.2	1.61	0	0.33	0	0.13	0	0	0	0	0.06	2.3	0	0.1	0.29	0.05	0	0	0.69	0.1	1.76	12.35
	DK	0	0.1	0	0	0	0.08	2.08	0	0.02	0.23	0.95	0	0	0.01	0	0	0	0	0	0.18	0.54	0	0.01	0	0	0	0.19	0.67	0.4	1.76	7.22
	EE	0	0.01	0	0	0	0.02	0.03	1.26	0.17	0.03	0.16	0	0	0	0	0.15	0.08	0	0	0.02	0.32	0	0.01	0	0	0	0.07	0.89	0.13	2.21	5.56
	FI	0	0	0	0	0	0.01	0.02	0.11	2.03	0.02	0.08	0	0	0	0	0.03	0.02	0	0	0.01	0.14	0	0	0	0	0	0.09	0.51	0.09	2.03	5.19
	FR	0.02	0.35	0	0	0	0.07	0	0	0	4.95	0.88	0	0	0.02	0.17	0	0	0.02	0	0.19	0.12	0	0	0	0	0.16	0	0.55	0.43	1.76	9.69
	DE	0.16	0.32	0	0	0	0.35	0.04	0	0	0.67	4.8	0	0.02	0.01	0.08	0	0	0.03	0	0.43	0.59	0	0	0.01	0	0.02	0.01	0.6	0.25	1.55	9.94
	GR	0	0	0.53	0	0	0.02	0	0	0	0.01	0.05	6.62	0.01	0	0.16	0	0	0	0	0	0.11	0	0.2	0.01	0	0.02	0	2.06	0.77	5.65	16.22
	HU	0.25	0.01	0.08	0.25	0	0.47	0	0	0	0.09	0.52	0.01	5.7	0	0.31	0	0	0	0	0.01	1.23	0	0.65	0.7	0.09	0	0	2.34	0.11	2.06	14.88
	IE	0	0.04	0	0	0	0.01	0	0	0	0.15	0.15	0	0	1.63	0	0	0	0	0	0.06	0.04	0	0	0	0	0.01	0	0.91	0.36	1.94	5.3
	IT	0.14	0	0.01	0.09	0	0.07	0	0	0	0.25	0.2	0.01	0.03	0	11.09	0	0	0	0	0	0.15	0	0.02	0.02	0.14	0.1	0	0.52	0.8	2.43	16.07
	LV	0	0.02	0	0	0	0.06	0.04	0.1	0.06	0.05	0.28	0	0.02	0	0	2.13	0.36	0	0	0.03	0.78	0	0.04	0.01	0	0	0.06	1.13	0.13	2.13	7.43
	LT	0.01	0.02	0	0	0	0.1	0.05	0.03	0.04	0.07	0.36	0	0.04	0	0.01	0.13	1.95	0	0	0.04	1.52	0	0.07	0.04	0	0	0.05	1.57	0.15	2.07	8.32
	LU	0.04	0.78	0	0	0	0.18	0	0	0	2.03	2.49	0	0	0.01	0.06	0	0	1.66	0	0.39	0.26	0	0	0	0	0.04	0	0.55	0.24	1.4	10.13
	MT	0.01	0	0.02	0.01	0	0.02	0	0	0	0.17	0.05	0.03	0	0	0.97	0	0	0	0.63	0	0.05	0.01	0.01	0	0	0.18	0	0.4	2.69	9.01	14.26
	NL	0	0.89	0	0	0	0.12	0.05	0	0	1.02	2.16	0	0	0.05	0.03	0	0	0.01	0	2.17	0.36	0	0	0	0	0.03	0	1.09	0.57	1.62	10.17
PL	0.07	0.03	0	0.01	0	0.61	0.04	0	0	0.12	0.9	0	0.21	0	0.07	0	0.02	0	0	0.05	10.16	0	0.13	0.26	0	0	0.01	1.07	0.13	1.94	15.83	
PT	0	0	0	0	0	0	0	0	0	0.1	0.02	0	0	0	0	0	0	0	0	0	0	4.67	0	0	0	1.15	0	0.03	0.8	2.93	9.7	
RO	0.02	0	0.38	0.01	0	0.08	0	0	0	0.01	0.16	0.02	0.24	0	0.08	0	0	0	0	0	0.52	0	7.84	0.09	0	0	0	2.44	0.14	2.77	14.8	
SK	0.25	0.01	0.03	0.11	0	0.74	0	0	0	0.09	0.56	0	1.09	0	0.19	0	0	0	0	0.02	2.07	0	0.32	2.96	0.06	0	1.24	0.08	1.99	11.81		
SI	0.61	0.01	0.01	0.65	0	0.28	0	0	0	0.11	0.48	0	0.28	0	1.86	0	0	0	0	0.01	0.49	0	0.11	0.08	4.09	0.02	0	0.82	0.27	1.65	11.83	
ES	0	0.01	0	0	0	0	0	0	0	0.33	0.07	0	0	0	0.07	0	0	0	0	0	0.01	0.16	0	0	0	5.92	0	0.04	0.96	3.14	10.71	
SE	0	0.03	0	0	0	0.03	0.19	0.01	0.06	0.07	0.31	0	0	0	0	0	0.01	0.01	0	0	0.05	0.3	0	0	0	0	0	0.99	0.45	0.19	1.79	4.49

Table 3.29: Origin of population-weighted PM_{2.5} background concentrations of PM2.5 in each Member State computed for 2015 (as percent of total background concentrations)

		Source regions																											Other Europe	Intern. shipping	Natural + Africa
		AT	BE	BG	HR	CY	CZ	DK	EE	FI	FR	DE	GR	HU	IE	IT	LV	LT	LU	MT	NL	PL	PT	RO	SK	SI	ES	SE			
Receptor regions	AT	34.1%	0.3%	0.1%	1.2%	0.0%	5.3%	0.0%	0.0%	0.0%	1.7%	12.8%	0.0%	2.8%	0.0%	5.1%	0.0%	0.0%	0.0%	0.0%	0.3%	6.7%	0.0%	0.7%	1.1%	2.5%	0.1%	0.0%	6.4%	0.9%	17.8%
	BE	0.0%	32.5%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%	15.9%	15.1%	0.0%	0.0%	0.4%	0.2%	0.0%	0.0%	0.3%	0.0%	7.4%	2.1%	0.0%	0.0%	0.0%	0.3%	0.0%	8.3%	3.9%	12.7%	
	BG	0.1%	0.0%	42.0%	0.0%	0.0%	0.4%	0.0%	0.0%	0.0%	0.0%	0.7%	1.3%	0.6%	0.0%	0.4%	0.0%	0.0%	0.0%	0.0%	0.0%	1.6%	0.0%	6.2%	0.2%	0.0%	0.0%	0.0%	19.0%	1.2%	26.3%
	HR	2.1%	0.1%	0.2%	24.8%	0.0%	2.3%	0.0%	0.0%	0.0%	0.6%	3.1%	0.0%	4.0%	0.0%	6.6%	0.0%	0.0%	0.0%	0.0%	0.1%	4.2%	0.0%	1.5%	1.0%	2.7%	0.2%	0.0%	27.4%	1.9%	17.4%
	CY	0.0%	0.0%	0.3%	0.0%	7.5%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	1.1%	0.0%	0.0%	0.2%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%	0.2%	0.0%	0.0%	0.0%	0.0%	28.9%	7.3%	54.3%
	CZ	2.6%	0.4%	0.1%	0.6%	0.0%	34.7%	0.0%	0.0%	0.0%	1.6%	13.0%	0.0%	2.7%	0.0%	1.1%	0.0%	0.0%	0.0%	0.0%	0.5%	18.6%	0.0%	0.8%	2.3%	0.4%	0.0%	0.0%	5.6%	0.8%	14.3%
	DK	0.0%	1.4%	0.0%	0.0%	0.0%	1.1%	28.8%	0.0%	0.3%	3.2%	13.2%	0.0%	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	2.5%	7.5%	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	2.6%	9.3%	5.5%	24.4%
	EE	0.0%	0.2%	0.0%	0.0%	0.0%	0.4%	0.5%	22.7%	3.1%	0.5%	2.9%	0.0%	0.0%	0.0%	0.0%	2.7%	1.4%	0.0%	0.0%	0.4%	5.8%	0.0%	0.2%	0.0%	0.0%	0.3%	1.3%	16.0%	2.3%	39.7%
	FI	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.4%	2.1%	39.1%	0.4%	1.5%	0.0%	0.0%	0.0%	0.0%	0.6%	0.4%	0.0%	0.0%	0.2%	2.7%	0.0%	0.0%	0.0%	0.0%	0.0%	1.7%	9.8%	1.7%	39.1%
	FR	0.2%	3.6%	0.0%	0.0%	0.0%	0.7%	0.0%	0.0%	0.0%	51.1%	9.1%	0.0%	0.0%	0.2%	1.8%	0.0%	0.0%	0.2%	0.0%	2.0%	1.2%	0.0%	0.0%	0.0%	0.0%	1.7%	0.0%	5.7%	4.4%	18.2%
	DE	1.6%	3.2%	0.0%	0.0%	0.0%	3.5%	0.4%	0.0%	0.0%	6.7%	48.3%	0.0%	0.2%	0.1%	0.8%	0.0%	0.0%	0.3%	0.0%	4.3%	5.9%	0.0%	0.0%	0.1%	0.0%	0.2%	0.1%	6.0%	2.5%	15.6%
	GR	0.0%	0.0%	3.3%	0.0%	0.0%	0.1%	0.0%	0.0%	0.0%	0.1%	0.3%	40.8%	0.1%	0.0%	1.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.7%	0.0%	1.2%	0.1%	0.0%	0.1%	0.0%	12.7%	4.7%	34.8%
	HU	1.7%	0.1%	0.5%	1.7%	0.0%	3.2%	0.0%	0.0%	0.0%	0.6%	3.5%	0.1%	38.3%	0.0%	2.1%	0.0%	0.0%	0.0%	0.1%	8.3%	0.0%	4.4%	4.7%	0.6%	0.0%	0.0%	15.7%	0.7%	13.8%	
	IE	0.0%	0.8%	0.0%	0.0%	0.0%	0.2%	0.0%	0.0%	0.0%	2.8%	2.8%	0.0%	0.0%	30.8%	0.0%	0.0%	0.0%	0.0%	1.1%	0.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.0%	17.2%	6.8%	36.6%
	IT	0.9%	0.0%	0.1%	0.6%	0.0%	0.4%	0.0%	0.0%	0.0%	1.6%	1.2%	0.1%	0.2%	0.0%	69.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.9%	0.0%	0.1%	0.1%	0.9%	0.6%	0.0%	3.2%	5.0%	15.1%
	LV	0.0%	0.3%	0.0%	0.0%	0.0%	0.8%	0.5%	1.3%	0.8%	0.7%	3.8%	0.0%	0.3%	0.0%	0.0%	28.7%	4.8%	0.0%	0.4%	10.5%	0.0%	0.5%	0.1%	0.0%	0.0%	0.8%	15.2%	1.7%	28.7%	
	LT	0.1%	0.2%	0.0%	0.0%	0.0%	1.2%	0.6%	0.4%	0.5%	0.8%	4.3%	0.0%	0.5%	0.0%	0.1%	1.6%	23.4%	0.0%	0.5%	18.3%	0.0%	0.8%	0.5%	0.0%	0.0%	0.6%	18.9%	1.8%	24.9%	
	LU	0.4%	7.7%	0.0%	0.0%	0.0%	1.8%	0.0%	0.0%	0.0%	20.0%	24.6%	0.0%	0.0%	0.1%	0.6%	0.0%	0.0%	16.4%	0.0%	3.8%	2.6%	0.0%	0.0%	0.0%	0.4%	0.0%	5.4%	2.4%	13.8%	
	MT	0.1%	0.0%	0.1%	0.1%	0.0%	0.1%	0.0%	0.0%	0.0%	1.2%	0.4%	0.2%	0.0%	0.0%	6.8%	0.0%	0.0%	0.0%	4.4%	0.0%	0.4%	0.1%	0.1%	0.0%	0.0%	1.3%	0.0%	2.8%	18.9%	63.2%
	NL	0.0%	8.8%	0.0%	0.0%	0.0%	1.2%	0.5%	0.0%	0.0%	10.0%	21.2%	0.0%	0.0%	0.5%	0.3%	0.0%	0.0%	0.1%	0.0%	21.3%	3.5%	0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	10.7%	5.6%	15.9%
PL	0.4%	0.2%	0.0%	0.1%	0.0%	3.9%	0.3%	0.0%	0.0%	0.8%	5.7%	0.0%	1.3%	0.0%	0.4%	0.0%	0.1%	0.0%	0.3%	64.2%	0.0%	0.8%	1.6%	0.0%	0.0%	0.1%	6.8%	0.8%	12.3%		
PT	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	1.0%	0.2%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	48.1%	0.0%	0.0%	0.0%	11.9%	0.0%	0.3%	8.2%	30.2%		
RO	0.1%	0.0%	2.6%	0.1%	0.0%	0.5%	0.0%	0.0%	0.0%	0.1%	1.1%	0.1%	1.6%	0.0%	0.5%	0.0%	0.0%	0.0%	0.0%	3.5%	0.0%	53.0%	0.6%	0.0%	0.0%	0.0%	16.5%	0.9%	18.7%		
SK	2.1%	0.1%	0.3%	0.9%	0.0%	6.3%	0.0%	0.0%	0.0%	0.8%	4.7%	0.0%	9.2%	0.0%	1.6%	0.0%	0.0%	0.0%	0.2%	17.5%	0.0%	2.7%	25.1%	0.5%	0.0%	0.0%	10.5%	0.7%	16.9%		
SI	5.2%	0.1%	0.1%	5.5%	0.0%	2.4%	0.0%	0.0%	0.0%	0.9%	4.1%	0.0%	2.4%	0.0%	15.7%	0.0%	0.0%	0.0%	0.1%	4.1%	0.0%	0.9%	0.7%	34.6%	0.2%	0.0%	6.9%	2.3%	13.9%		
ES	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.1%	0.7%	0.0%	0.0%	0.0%	0.7%	0.0%	0.0%	0.0%	0.0%	0.1%	1.5%	0.0%	0.0%	0.0%	55.3%	0.0%	0.4%	9.0%	29.3%		
SE	0.0%	0.7%	0.0%	0.0%	0.0%	0.7%	4.2%	0.2%	1.3%	1.6%	6.9%	0.0%	0.0%	0.0%	0.0%	0.2%	0.2%	0.0%	1.1%	6.7%	0.0%	0.0%	0.0%	0.0%	0.0%	22.0%	10.0%	4.2%	39.9%		

Table 3.30: Origin of population-weighted PM_{2.5} concentrations in each Member State computed for the CAO2 baseline scenario in 2030 (µg/m³)

		Source regions																												Other Europe	Intern. shipping	Natural + Africa	Total
		AT	BE	BG	HR	CY	CZ	DK	EE	FI	FR	DE	GR	HU	IE	IT	LV	LT	LU	MT	NL	PL	PT	RO	SK	SI	ES	SE					
Receptor regions	AT	1.82	0.02	0	0.07	0	0.28	0	0	0	0.1	0.8	0	0.14	0	0.31	0	0	0	0	0.02	0.3	0	0.03	0.05	0.15	0	0	0.33	0.08	1.73	6.23	
	BE	0	3.15	0	0	0	0.05	0	0	0	1.24	1.27	0	0	0.03	0.02	0	0	0.03	0	0.71	0.13	0	0	0	0	0.01	0	0.56	0.44	1.57	9.21	
	BG	0	0	2.42	0	0	0.02	0	0	0	0	0.07	0.1	0.04	0	0.04	0	0	0	0	0	0.09	0	0.4	0.01	0	0	0	1.36	0.1	3.52	8.17	
	HR	0.17	0.01	0.01	1.12	0	0.16	0	0	0	0.05	0.26	0	0.25	0	0.58	0	0	0	0	0	0.25	0	0.1	0.05	0.21	0.01	0	1.43	0.16	2.27	7.09	
	CY	0	0	0.01	0	0.53	0	0	0	0	0	0.01	0.06	0	0	0.03	0	0	0	0	0	0	0	0.01	0	0	0	0	2.73	0.56	7.02	10.96	
	CZ	0.2	0.03	0	0.05	0	1.82	0	0	0	0.12	1.1	0	0.17	0	0.09	0	0	0	0	0	0.05	1.17	0	0.05	0.13	0.04	0	0	0.35	0.09	1.76	7.22
	DK	0	0.07	0	0	0	0.04	1.04	0	0.01	0.14	0.66	0	0	0.01	0	0	0	0	0	0.13	0.31	0	0	0	0	0	0.15	0.41	0.38	1.76	5.11	
	EE	0	0.01	0	0	0	0.01	0.02	0.69	0.12	0.02	0.11	0	0	0	0	0.1	0.05	0	0	0.02	0.17	0	0.01	0	0	0	0.05	0.82	0.14	2.21	4.55	
	FI	0	0	0	0	0	0	0.01	0.07	1.38	0.01	0.06	0	0	0	0	0.02	0.02	0	0	0.01	0.07	0	0	0	0	0	0.06	0.46	0.09	2.03	4.29	
	FR	0.01	0.27	0	0	0	0.04	0	0	0	2.76	0.59	0	0	0.01	0.12	0	0	0.02	0	0.14	0.06	0	0	0	0	0.08	0	0.33	0.31	1.76	6.5	
	DE	0.1	0.25	0	0	0	0.19	0.02	0	0	0.41	3.41	0	0.01	0.01	0.05	0	0	0.02	0	0.33	0.31	0	0	0	0	0.01	0	0.38	0.22	1.55	7.27	
	GR	0	0	0.23	0	0	0.01	0	0	0	0.01	0.03	2.93	0.01	0	0.1	0	0	0	0	0	0.04	0	0.09	0	0	0.01	0	1.12	0.39	5.65	10.62	
	HU	0.16	0.01	0.04	0.15	0	0.26	0	0	0	0.05	0.34	0	1.92	0	0.21	0	0	0	0	0.01	0.61	0	0.31	0.31	0.06	0	0	1.13	0.08	2.06	7.71	
	IE	0	0.03	0	0	0	0	0	0	0	0.1	0.1	0	0	0.98	0	0	0	0	0	0.04	0.02	0	0	0	0	0	0	0.5	0.23	1.94	3.94	
	IT	0.09	0	0	0.05	0	0.04	0	0	0	0.15	0.13	0	0.02	0	5.9	0	0	0	0	0	0.06	0	0.01	0	0.09	0.05	0	0.21	0.47	2.43	9.7	
	LV	0	0.01	0	0	0	0.03	0.03	0.06	0.04	0.03	0.19	0	0.01	0	0	1.07	0.23	0	0	0.02	0.44	0	0.02	0	0	0	0.04	1.02	0.12	2.13	5.49	
	LT	0	0.02	0	0	0	0.06	0.03	0.02	0.02	0.04	0.24	0	0.02	0	0	0.08	0.77	0	0	0.03	0.84	0	0.04	0.02	0	0	0.03	1.39	0.14	2.07	5.86	
	LU	0.03	0.61	0	0	0	0.1	0	0	0	1.24	1.73	0	0	0.01	0.04	0	0	1.6	0	0.29	0.13	0	0	0	0	0.02	0	0.32	0.21	1.4	7.73	
	MT	0	0	0.01	0.01	0	0.01	0	0	0	0.11	0.04	0.01	0	0	0.63	0	0	0	0.41	0	0.02	0	0	0	0	0	0.1	0	0.1	1.41	9.01	11.87
	NL	0	0.7	0	0	0	0.06	0.03	0	0	0.63	1.44	0	0	0.03	0.02	0	0	0.01	0	1.82	0.18	0	0	0	0	0.01	0	0.6	0.54	1.62	7.69	
	PL	0.05	0.02	0	0.01	0	0.33	0.03	0	0	0.07	0.61	0	0.11	0	0.05	0	0.01	0	0	0.04	4.77	0	0.07	0.12	0	0	0.01	0.74	0.12	1.94	9.1	
	PT	0	0	0	0	0	0	0	0	0	0.07	0.02	0	0	0	0	0	0	0	0	0	0	2.08	0	0	0	0.59	0	0.01	0.43	2.93	6.13	
	RO	0.01	0	0.2	0.01	0	0.04	0	0	0	0.01	0.1	0.01	0.12	0	0.05	0	0	0	0	0	0.24	0	2.85	0.03	0	0	0	1.42	0.09	2.77	7.95	
	SK	0.16	0.01	0.02	0.07	0	0.41	0	0	0	0.05	0.38	0	0.56	0	0.12	0	0	0	0	0.01	1.08	0	0.16	1.17	0.04	0	0	0.66	0.06	1.99	6.95	
	SI	0.4	0	0	0.36	0	0.15	0	0	0	0.07	0.31	0	0.13	0	1.26	0	0	0	0	0	0.22	0	0.05	0.03	2.02	0.01	0	0.29	0.19	1.65	7.14	
	ES	0	0.01	0	0	0	0	0	0	0	0.21	0.05	0	0	0	0.05	0	0	0	0	0	0	0.09	0	0	0	2.81	0	0.02	0.49	3.14	6.87	
	SE	0	0.02	0	0	0	0.02	0.13	0.01	0.04	0.05	0.22	0	0	0	0	0	0.01	0	0	0.04	0.16	0	0	0	0	0	0.74	0.34	0.18	1.79	3.75	

Table 3.31: Origin of population-weighted PM_{2.5} background concentrations of PM2.5 in each Member State computed for the CAO2 baseline scenario in 2030 (as percent of total background concentrations)

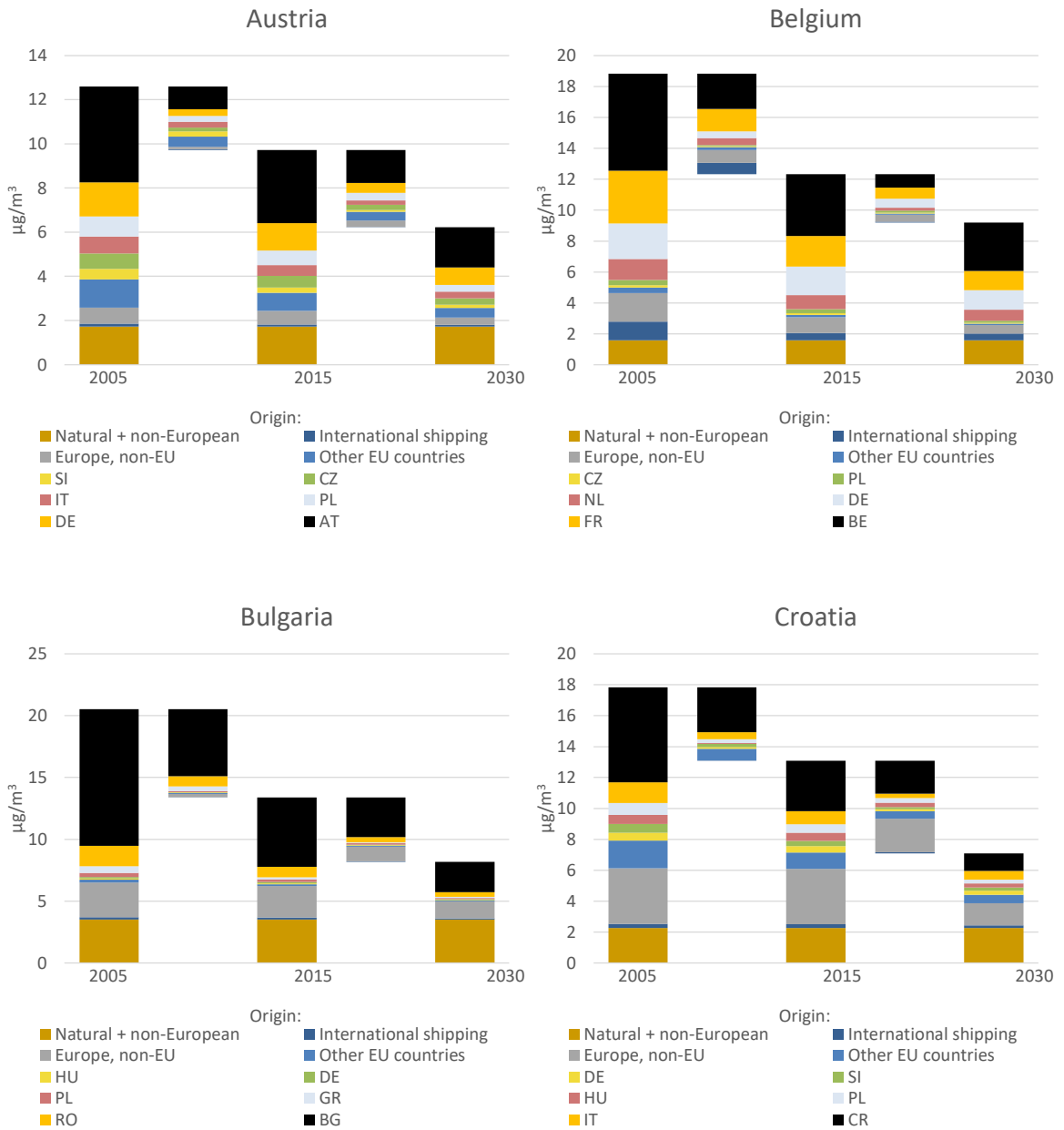
		Source regions																											Other Europe	Intern. shipping	Natural + Africa
		AT	BE	BG	HR	CY	CZ	DK	EE	FI	FR	DE	GR	HU	IE	IT	LV	LT	LU	MT	NL	PL	PT	RO	SK	SI	ES	SE			
Receptor regions	AT	29.2%	0.3%	0.0%	1.1%	0.0%	4.5%	0.0%	0.0%	0.0%	1.6%	12.8%	0.0%	2.2%	0.0%	5.0%	0.0%	0.0%	0.0%	0.0%	0.3%	4.8%	0.0%	0.5%	0.8%	2.4%	0.0%	0.0%	5.3%	1.3%	27.8%
	BE	0.0%	34.2%	0.0%	0.0%	0.0%	0.5%	0.0%	0.0%	0.0%	13.5%	13.8%	0.0%	0.0%	0.3%	0.2%	0.0%	0.0%	0.3%	0.0%	7.7%	1.4%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%	6.1%	4.8%	17.0%
	BG	0.0%	0.0%	29.6%	0.0%	0.0%	0.2%	0.0%	0.0%	0.0%	0.0%	0.9%	1.2%	0.5%	0.0%	0.5%	0.0%	0.0%	0.0%	0.0%	0.0%	1.1%	0.0%	4.9%	0.1%	0.0%	0.0%	0.0%	16.6%	1.2%	43.1%
	HR	2.4%	0.1%	0.1%	15.8%	0.0%	2.3%	0.0%	0.0%	0.0%	0.7%	3.7%	0.0%	3.5%	0.0%	8.2%	0.0%	0.0%	0.0%	0.0%	0.0%	3.5%	0.0%	1.4%	0.7%	3.0%	0.1%	0.0%	20.2%	2.3%	32.0%
	CY	0.0%	0.0%	0.1%	0.0%	4.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.5%	0.0%	0.0%	0.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	24.9%	5.1%	64.1%
	CZ	2.8%	0.4%	0.0%	0.7%	0.0%	25.2%	0.0%	0.0%	0.0%	1.7%	15.2%	0.0%	2.4%	0.0%	1.2%	0.0%	0.0%	0.0%	0.0%	0.7%	16.2%	0.0%	0.7%	1.8%	0.6%	0.0%	0.0%	4.8%	1.2%	24.4%
	DK	0.0%	1.4%	0.0%	0.0%	0.0%	0.8%	20.4%	0.0%	0.2%	2.7%	12.9%	0.0%	0.0%	0.2%	0.0%	0.0%	0.0%	0.0%	0.0%	2.5%	6.1%	0.0%	0.0%	0.0%	0.0%	0.0%	2.9%	8.0%	7.4%	34.4%
	EE	0.0%	0.2%	0.0%	0.0%	0.0%	0.2%	0.4%	15.2%	2.6%	0.4%	2.4%	0.0%	0.0%	0.0%	0.0%	2.2%	1.1%	0.0%	0.0%	0.4%	3.7%	0.0%	0.2%	0.0%	0.0%	0.0%	1.1%	18.0%	3.1%	48.6%
	FI	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	1.6%	32.2%	0.2%	1.4%	0.0%	0.0%	0.0%	0.0%	0.5%	0.5%	0.0%	0.0%	0.2%	1.6%	0.0%	0.0%	0.0%	0.0%	0.0%	1.4%	10.7%	2.1%	47.3%
	FR	0.2%	4.2%	0.0%	0.0%	0.0%	0.6%	0.0%	0.0%	0.0%	42.5%	9.1%	0.0%	0.0%	0.2%	1.8%	0.0%	0.0%	0.3%	0.0%	2.2%	0.9%	0.0%	0.0%	0.0%	0.0%	1.2%	0.0%	5.1%	4.8%	27.1%
	DE	1.4%	3.4%	0.0%	0.0%	0.0%	2.6%	0.3%	0.0%	0.0%	5.6%	46.9%	0.0%	0.1%	0.1%	0.7%	0.0%	0.0%	0.3%	0.0%	4.5%	4.3%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%	5.2%	3.0%	21.3%
	GR	0.0%	0.0%	2.2%	0.0%	0.0%	0.1%	0.0%	0.0%	0.0%	0.1%	0.3%	27.6%	0.1%	0.0%	0.9%	0.0%	0.0%	0.0%	0.0%	0.0%	0.4%	0.0%	0.8%	0.0%	0.0%	0.1%	0.0%	10.5%	3.7%	53.2%
	HU	2.1%	0.1%	0.5%	1.9%	0.0%	3.4%	0.0%	0.0%	0.0%	0.6%	4.4%	0.0%	24.9%	0.0%	2.7%	0.0%	0.0%	0.0%	0.0%	0.1%	7.9%	0.0%	4.0%	4.0%	0.8%	0.0%	0.0%	14.7%	1.0%	26.7%
	IE	0.0%	0.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.5%	2.5%	0.0%	0.0%	24.9%	0.0%	0.0%	0.0%	0.0%	0.0%	1.0%	0.5%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	12.7%	5.8%	49.2%
	IT	0.9%	0.0%	0.0%	0.5%	0.0%	0.4%	0.0%	0.0%	0.0%	1.5%	1.3%	0.0%	0.2%	0.0%	60.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.6%	0.0%	0.1%	0.0%	0.9%	0.5%	0.0%	2.2%	4.8%	25.1%
	LV	0.0%	0.2%	0.0%	0.0%	0.0%	0.5%	0.5%	1.1%	0.7%	0.5%	3.5%	0.0%	0.2%	0.0%	0.0%	19.5%	4.2%	0.0%	0.0%	0.4%	8.0%	0.0%	0.4%	0.0%	0.0%	0.0%	0.7%	18.6%	2.2%	38.8%
	LT	0.0%	0.3%	0.0%	0.0%	0.0%	1.0%	0.5%	0.3%	0.3%	0.7%	4.1%	0.0%	0.3%	0.0%	0.0%	1.4%	13.1%	0.0%	0.0%	0.5%	14.3%	0.0%	0.7%	0.3%	0.0%	0.0%	0.5%	23.7%	2.4%	35.3%
LU	0.4%	7.9%	0.0%	0.0%	0.0%	1.3%	0.0%	0.0%	0.0%	16.0%	22.4%	0.0%	0.0%	0.1%	0.5%	0.0%	0.0%	20.7%	0.0%	3.8%	1.7%	0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	4.1%	2.7%	18.1%	
MT	0.0%	0.0%	0.1%	0.1%	0.0%	0.1%	0.0%	0.0%	0.0%	0.9%	0.3%	0.1%	0.0%	0.0%	5.3%	0.0%	0.0%	0.0%	3.5%	0.0%	0.2%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.8%	11.9%	75.9%	
NL	0.0%	9.1%	0.0%	0.0%	0.0%	0.8%	0.4%	0.0%	0.0%	8.2%	18.7%	0.0%	0.0%	0.4%	0.3%	0.0%	0.0%	0.1%	0.0%	23.7%	2.3%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%	7.8%	7.0%	21.1%	
PL	0.5%	0.2%	0.0%	0.1%	0.0%	3.6%	0.3%	0.0%	0.0%	0.8%	6.7%	0.0%	1.2%	0.0%	0.5%	0.0%	0.1%	0.0%	0.0%	0.4%	52.4%	0.0%	0.8%	1.3%	0.0%	0.0%	0.1%	8.1%	1.3%	21.3%	
PT	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	1.1%	0.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	33.9%	0.0%	0.0%	0.0%	9.6%	0.0%	0.2%	7.0%	47.8%	
RO	0.1%	0.0%	2.5%	0.1%	0.0%	0.5%	0.0%	0.0%	0.0%	0.1%	1.3%	0.1%	1.5%	0.0%	0.6%	0.0%	0.0%	0.0%	0.0%	0.0%	3.0%	0.0%	35.8%	0.4%	0.0%	0.0%	0.0%	17.9%	1.1%	34.8%	
SK	2.3%	0.1%	0.3%	1.0%	0.0%	5.9%	0.0%	0.0%	0.0%	0.7%	5.5%	0.0%	8.1%	0.0%	1.7%	0.0%	0.0%	0.0%	0.0%	0.1%	15.5%	0.0%	2.3%	16.8%	0.6%	0.0%	0.0%	9.5%	0.9%	28.6%	
SI	5.6%	0.0%	0.0%	5.0%	0.0%	2.1%	0.0%	0.0%	0.0%	1.0%	4.3%	0.0%	1.8%	0.0%	17.6%	0.0%	0.0%	0.0%	0.0%	0.0%	3.1%	0.0%	0.7%	0.4%	28.3%	0.1%	0.0%	4.1%	2.7%	23.1%	
ES	0.0%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.1%	0.7%	0.0%	0.0%	0.0%	0.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	1.3%	0.0%	0.0%	0.0%	40.9%	0.0%	0.3%	7.1%	45.7%	
SE	0.0%	0.5%	0.0%	0.0%	0.0%	0.5%	3.5%	0.3%	1.1%	1.3%	5.9%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	0.0%	1.1%	4.3%	0.0%	0.0%	0.0%	0.0%	0.0%	19.7%	9.1%	4.8%	47.7%	

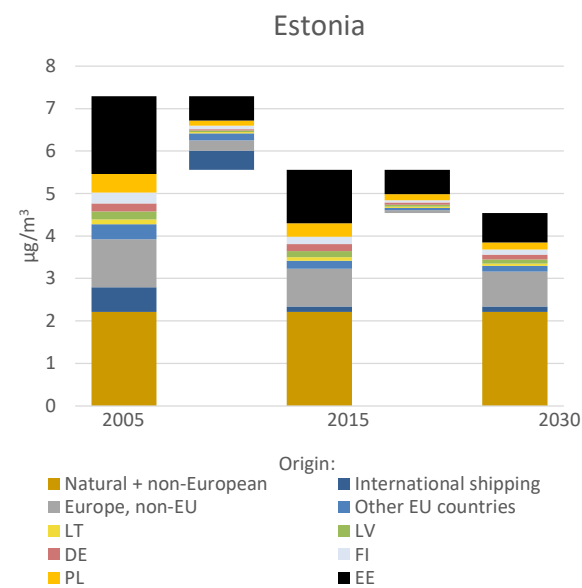
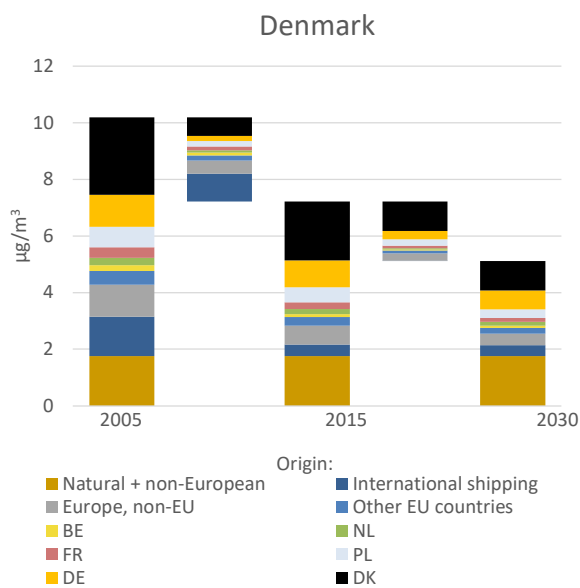
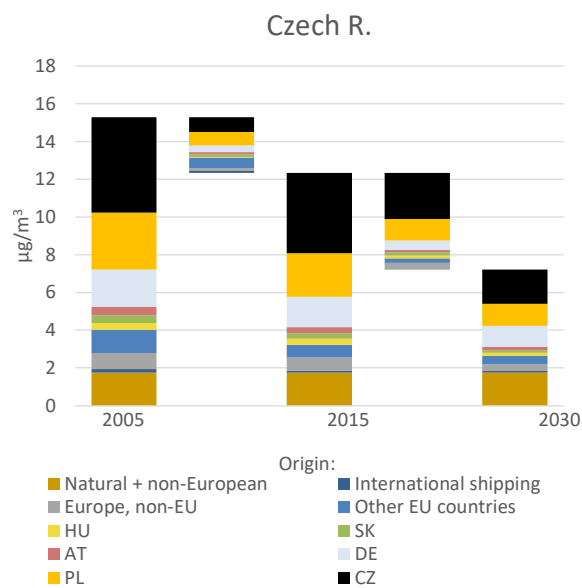
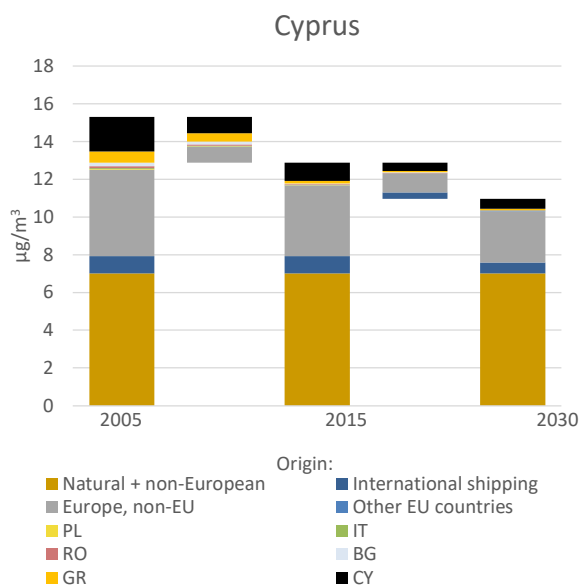
3.7.2 Origin of PM_{2.5} in each Member State

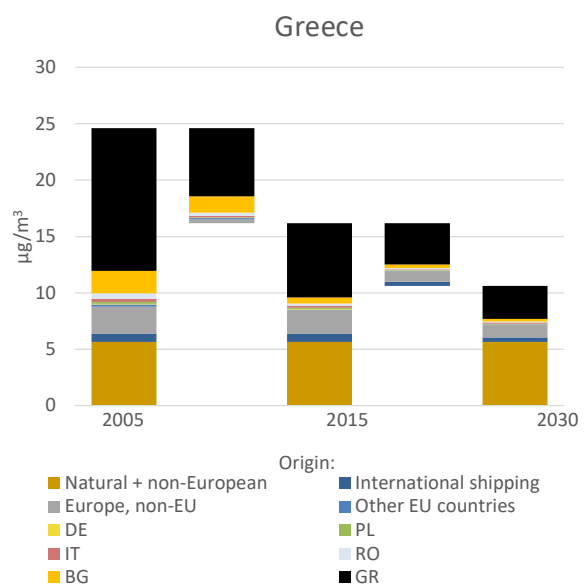
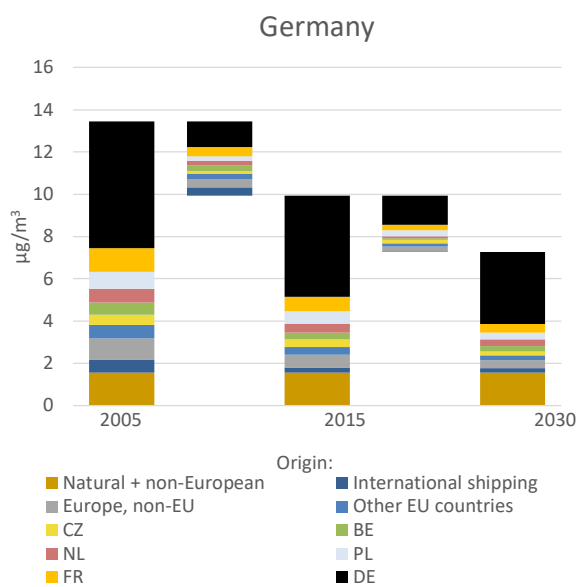
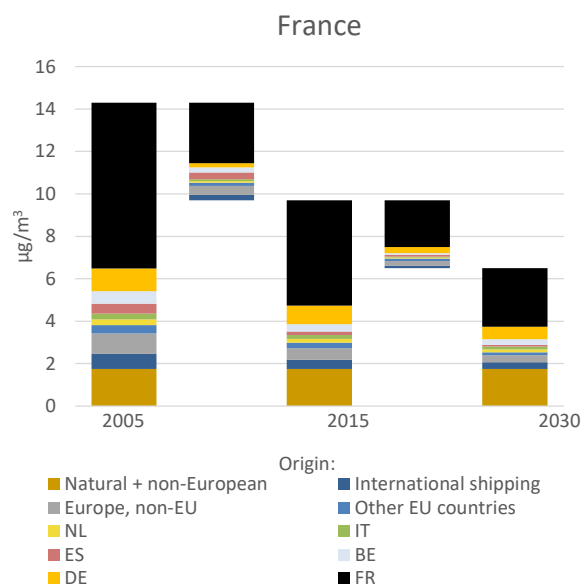
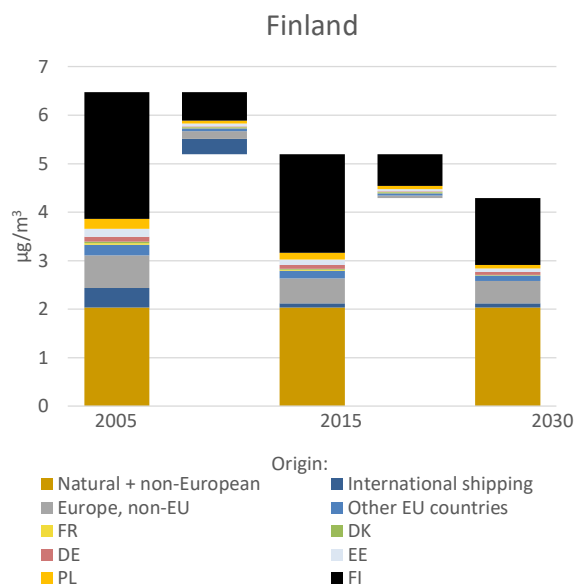
To reveal the importance of the international linkages in air quality management, and in particular the benefits of EU-wide coordinated emission reductions on the air quality in each country, the origins of the contributions to PM_{2.5} background concentrations in each Member State have been compared for different scenarios and years.

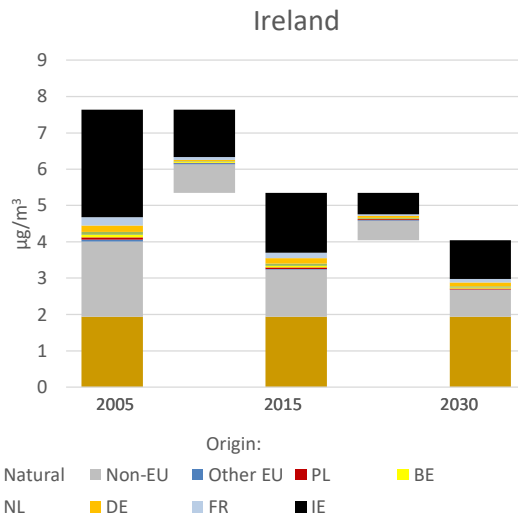
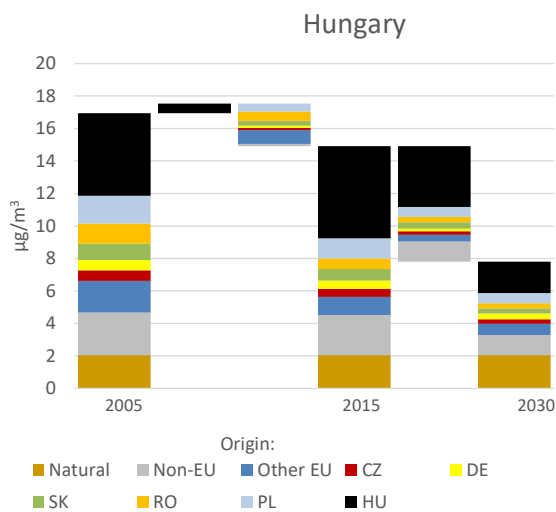
Figure 3.20 shows for all Member States the (changes) in the contributions to population-weighted PM_{2.5} exposure (for background concentration only) for 2005, 2015 and the baseline scenario in 2030. These show not only the significant impacts of the domestic emission reductions on air quality in each Member State, but also how EU policies through coordinated measures in all EU Member countries enhanced the positive effect by reducing the inflow of pollution coming from neighbouring countries. However, in some countries (Croatia, Malta, etc.) higher emissions from international shipping lead to higher contributions from outside the EU.

Figure 3.20: Origins of (population-weighted) ambient background concentrations of PM_{2.5} in each Member State in 2005, 2015 and 2030, and the contributions to the changes between (a) 2005 and 2015, and (b) 2015 and 2030. Note that the category ‘Europe, non-EU’ includes Belarus, Norway, the European part of Russia, Switzerland, Turkey, the UK and the Ukraine.

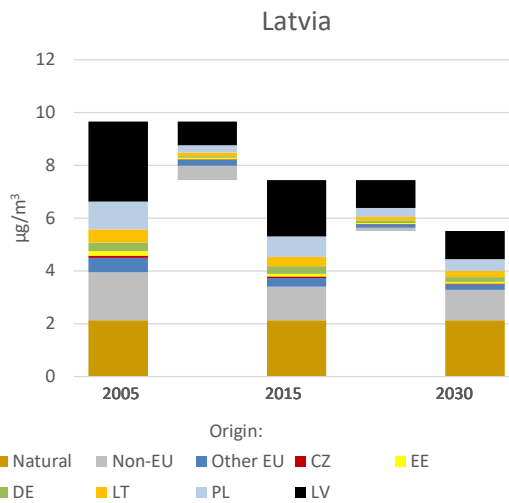
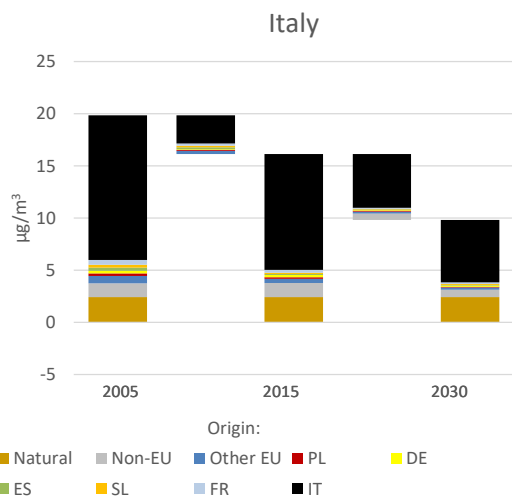


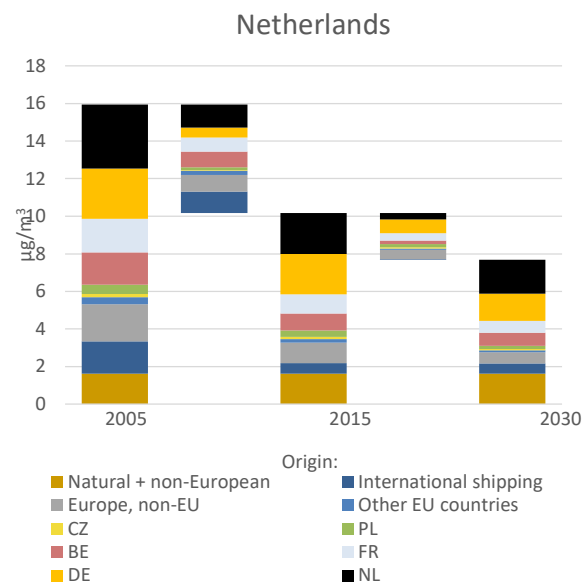
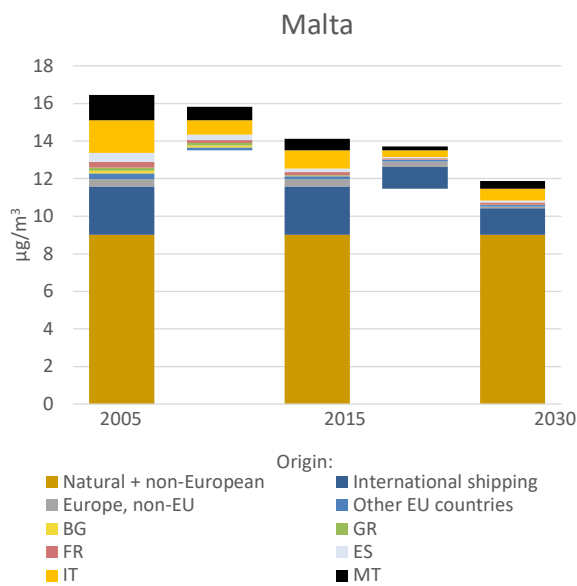
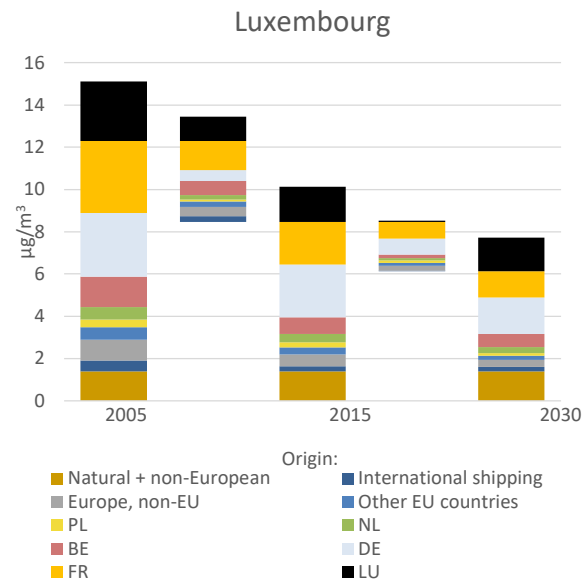
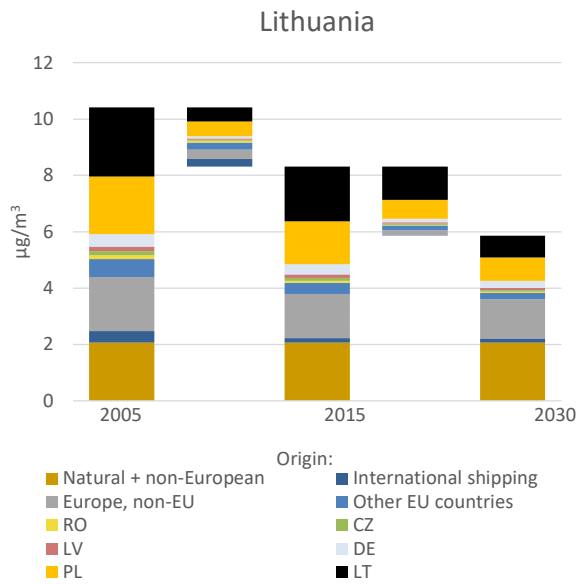


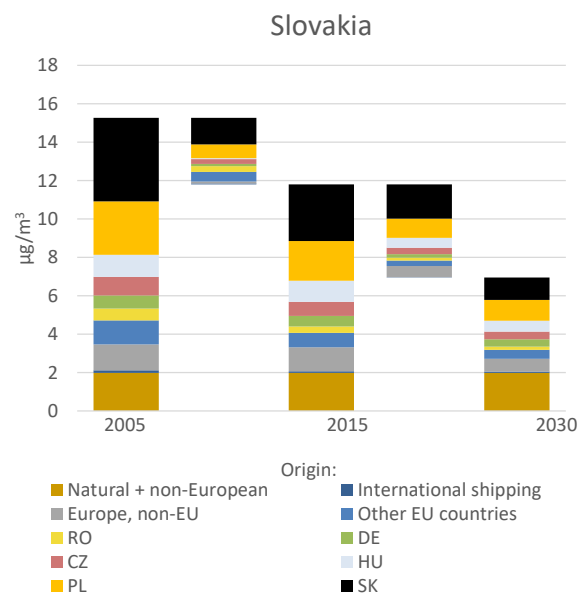
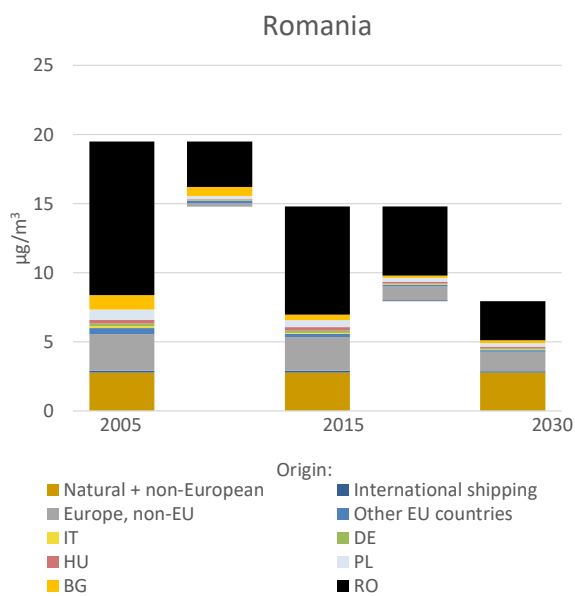
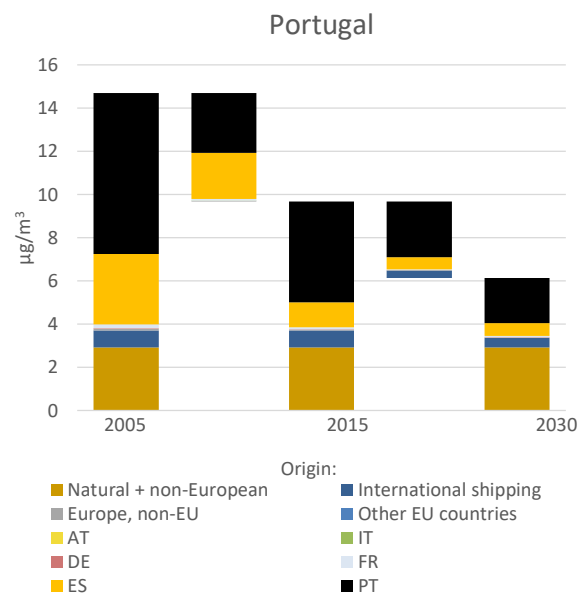
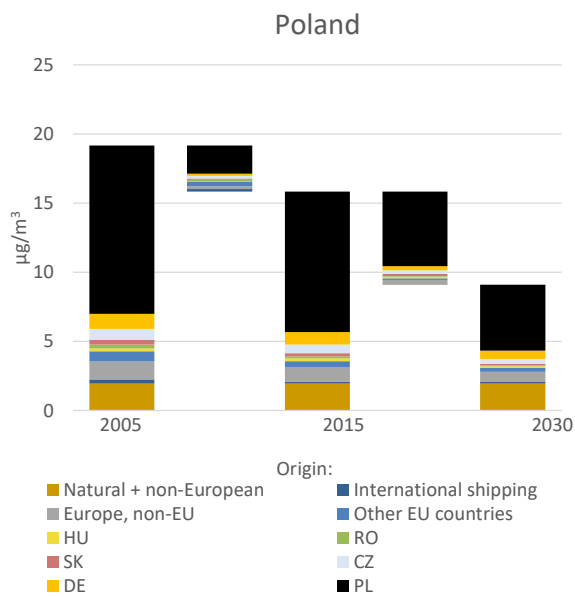


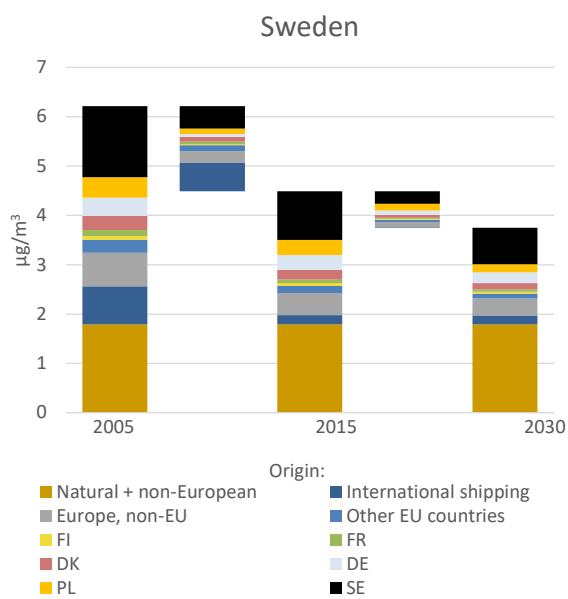
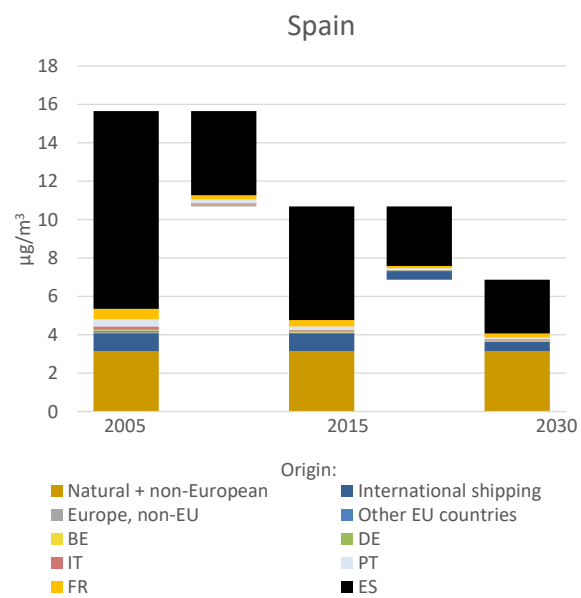
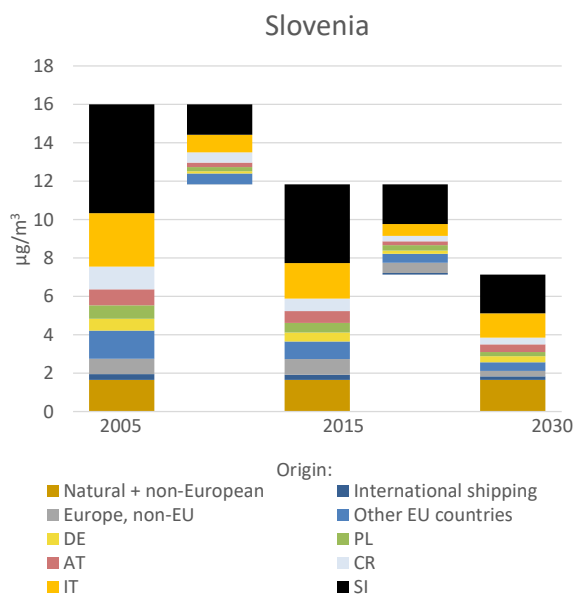


In Hungary, the domestic contribution increased between 2005 and 2015, while imported pollution from other countries declined.









3.7.3 Air quality benefits of emission controls on other countries

The import-export matrices can also be used to reveal the extent to which the benefits from emission control measures in a country occur within the same country, and how much of the total benefits materialise outside. This is of particular importance for national benefits analyses of domestic air pollution control policies and measures, which often ignore the benefits outside their own country and thereby provide only an incomplete account of benefits.

The analysis presented for this study quantifies the improvements in (cumulated) population exposure to ambient PM_{2.5} (background concentration only) resulting from emission reduction between 2015 and the 2030 baseline projection; it therefore allows to distinguish between the share of population exposure reductions within the same country and in the other EU Member States (Figure 3.21). The calculation takes into account the changes in precursor emissions between 2015 and 2030 baseline projection, the dispersion and transport of the PM_{2.5} precursor emissions in the atmosphere, how this influences ambient PM_{2.5} levels across the EU, and how many people are affected by these changes (using population figures for 2030).

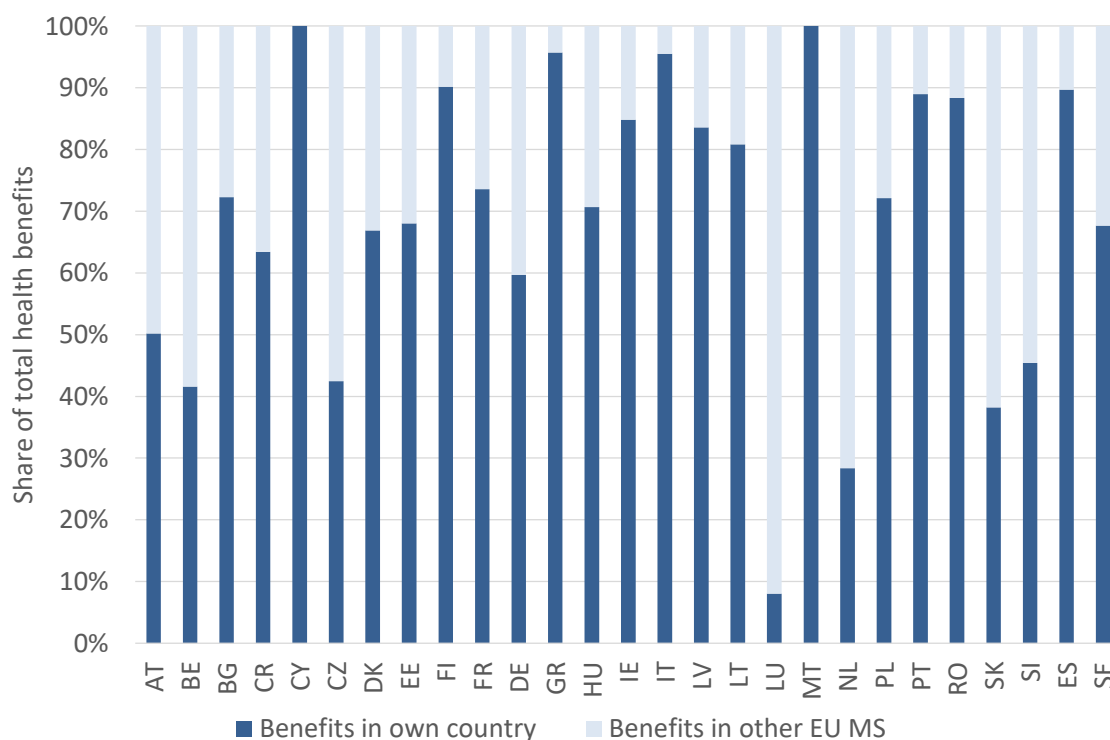


Figure 3.21: Share of the reduction in population exposure to PM_{2.5} (background concentration only) due to domestic emission reductions from 2015 to 2030 in the CAO2 baseline projections that occurred in the same country where emissions are reduced.

4 Key findings

This report presents an outlook on the likely evolution of air quality in the European Union and on associated health, ecosystems and economic impacts. To inform the development of the Second Clean Air Outlook, the model analysis has incorporated the most recent information provided by Member States on national emission inventories as well as on additional air pollution and climate policies and measures that are already selected for implementation. Also, the latest projections of energy and agricultural activities have been included. Health impact assessment methods and approaches for the economic valuation of benefits have been updated based on reviews of the latest scientific literature, indicating larger benefits from clean air than estimated previously.

The updated model framework clearly shows that, with full implementation of all currently existing sectoral legislation regulating air pollution and of the measures needed to reach the Climate and Energy targets for 2030 that were set in 2018, the NEC Directive requirements would be fulfilled at the EU level in 2030 for all pollutants except ammonia. However, this hides differences among Member States and in some cases even NAPCP measures would not be sufficient to reach the NEC Directive emission reduction commitments (especially in the case of ammonia).

In the same context, in all air quality management zones, background concentrations of PM_{2.5} would fall below the current limit value of 25 µg/m³. In 90 percent of all air quality management zones, background concentrations of PM_{2.5} would decline below the current WHO guidelines value of 10 µg/m³ (in 2015, background PM_{2.5} levels were below the guideline value in only 41% of the air quality management zones). Full implementation of all technically feasible air pollution control measures combined with further climate policies could bring background concentrations of PM_{2.5} below the WHO guideline in almost 99% of the zones.

The parallel decline in population exposure to PM_{2.5} will be mirrored by significantly lower health impacts by 2030. If all policies that Member States have already agreed to are fully implemented, it is estimated that the envisaged air quality improvements will save annually more than 210,000 cases of premature deaths attributable to PM_{2.5}, which accounts for about 55% of the number in 2005.

When taking into account the additional policies and measures reported by Member States in their NAPCP (only the ones which were defined with enough detail to be quantified though), around 7,000 additional premature deaths due to PM_{2.5} alone would be avoided annually in the EU, for each year between 2030 to 2050. These NAPCP measures would bring health benefits between EUR 8-43 billion per year (the range represents the variety of valuation assumptions) and always bring a net benefit to society. Full implementation of all technically feasible emission controls could save another 30,000 lives every year. However, this would still leave more than 130,000 premature deaths in the EU due to PM_{2.5} pollution alone, mostly due to pollution coming from outside of the EU including international maritime activities.

In addition to their health benefits, the envisaged emission reductions will also enhance the sustainability of ecosystems. However, excess nitrogen deposition, an important risk factor for biodiversity of terrestrial and aquatic ecosystems, will remain of high concern, as the critical loads for eutrophication will remain widely exceeded. The emission reduction requirements of the NEC Directive include NH₃ emissions, as an important precursor for eutrophication. Still, a large potential for further reduction through changes in agricultural practices and lifestyle changes (including human diets) will still remain. Such additional reductions could not only make important contributions towards the protection of biodiversity of the European ecosystems, especially in the Natura 2000 nature protection areas, but at the same time improve human health through lower PM_{2.5} levels and reduce emissions of greenhouse gases.

The model analyses conducted for this report clearly indicate that additional actions towards cleaner air always bring net benefits to society. Also, in general, climate policy measures deliver co-benefits on air quality, although an enhanced use of fuel wood in the residential sector can counteract clean air efforts through higher emissions of PM_{2.5}.

Due to the residence time of PM_{2.5} in the atmosphere air of several days, PM_{2.5} can be transported over several hundreds of kilometres. As a consequence, PM_{2.5} at any given location originates from sources in a large area, often in other countries. The high share of PM_{2.5} background concentration originating from other Member States and non-EU countries underlines the need for EU legislation and for the EU to take international action. In the past, such coordinated EU-wide action has shown the increased reductions in background PM_{2.5} concentrations compared to the impacts of domestic actions in the same country. Increasing marginal abatement costs call for even further EU-wide and international coordination, in order to realise the potentials for additional air quality improvements at least cost.

In the long run, with the further policies and measures that have been analysed in this report, it seems possible to bring background concentrations of PM_{2.5} below the current WHO guideline value of 10 µg/m³ throughout Europe. Possible exceptions remain at local hot spots that are not captured by the spatial resolution of this analysis.

References

- Aguiar, A., Chepeliev, M., Corong, E., McDougall, R., & van der Mensbrugghe, D. (2019). The GTAP Data Base: Version 10. *Journal of Global Economic Analysis*, 4(1), 1-27
- Amann, M. (2018). Progress towards the achievement of the EU's air quality and emissions objectives [IIASA report]. IIASA - International Institute for Applied Systems Analysis. https://ec.europa.eu/environment/air/pdf/clean_air_outlook_overview_report.pdf
- Chang, T., Graff Zivin, J., Gross, T., & Neidell, M. (2016). Particulate pollution and the productivity of pear packers. *American Economic Journal: Economic Policy*, 8(3), 141-69.
- Cofala, J., Amann, M., Borken-Kleefeld, J., Gomez Sanabria, A., Heyes, C., Kiesewetter, G., Sander, R., Schöpp, W., Holland, M., Fagerli, H., & Nyiri, A. (2018). The potential for cost-effective air emission reductions from international shipping through designation of further Emission Control Areas in EU waters with focus on the Mediterranean Sea [Final Report]. International Institute for Applied Systems Analysis (IIASA). <http://pure.iiasa.ac.at/id/eprint/15729/>
- Dechezleprêtre, A., Rivers, N., & Stadler, B. (2019). The economic cost of air pollution: Evidence from Europe. OECD Economics Department Working Papers
- Denier van der Gon, H. A. C., Bergström, R., Fountoukis, C., Johansson, C., Pandis, S. N., Simpson, D., & Visschedijk, A. J. H. (2015). Particulate emissions from residential wood combustion in Europe – revised estimates and an evaluation. *Atmospheric Chemistry and Physics*, 15(11), 6503–6519. <https://doi.org/10.5194/acp-15-6503-2015>
- EEA. (2016). EMEP/EEA air pollutant emission inventory guidebook 2016 (Technical Report No 21/2016). European Environmental Agency (EEA). <https://www.eea.europa.eu/publications/emep-eea-guidebook-2016>
- FAO, & ECE. (2018). Wood Energy in the ECE Region (ECE/TIM/SP/42). United Nations Economic Commission for Europe
- Gómez-Sanabria, A., Höglund-Isaksson, L., Rafaj, P., & Schöpp, W. (2018). Carbon in global waste and wastewater flows – its potential as energy source under alternative future waste management regimes. *Advances in Geosciences*, 45, 105–113. <https://doi.org/10.5194/adgeo-45-105-2018>
- HBEFA 4.1. (2019). HBEFA - Handbook Emission Factors for Road Transport. <http://www.hbefa.net/d/index.html>
- Hettelingh J-P, Posch M, Slootweg J (eds), 2017. European critical loads: database, biodiversity and ecosystems at risk: CCE Final Report 2017. RIVM Report 2017-0155, Coordination Centre for Effects, Bilthoven, Netherlands, 204 pp; ISBN 978-90-6960-288-2, DOI: 10.21945/RIVM-2017-0155
- IEA. (2018). World Energy Outlook 2018. International Energy Agency (IEA)
- Kahn, M. E., & Li, P. (2020). Air pollution lowers high skill public sector worker productivity in China. *Environmental Research Letters*, 15(8), 084003
- Klimont, Z., Cofala, J., Bertok, I., Amann, M., Heyes, C., & Gyarmas, F. (2002). Modelling Particulate Emissions in Europe. A Framework to Estimate Reduction Potential and Control Costs (Interim Report IR-02-076). International Institute for Applied Systems Analysis (IIASA)
- Klimont, Z., Kupiainen, K., Heyes, C., Purohit, P., Cofala, J., Rafaj, P., Borken-Kleefeld, J., & Schöpp, W. (2017). Global anthropogenic emissions of particulate matter including black carbon. *Atmospheric Chemistry and Physics*, 17(14), 8681–8723. <https://doi.org/10.5194/acp-17-8681-2017>
- Nussbaumer, T. (2010). Overview on Technologies for Biomass Combustion and Emission Levels of Particulate Matter (p. 78) [Prepared for Swiss Federal Office for the Environment]. Expert Group on Techno-Economic Issues

Randerson, J. T., van der Werf, G. R., Giglio, L., Collatz, G. J., & Kasibhatla, P. S. (2015). Global Fire Emissions Database, Version 4, (GFEDv4). ORNL Distributed Active Archive Center. <http://dx.doi.org/10.3334/ORNLDAAAC/1293>

WHO Regional Office for Europe. (2013). Health risks of air pollution in Europe – HRAPIE project. World Health Organization, Regional Office for Europe

Wiedinmyer, C., Akagi, S. K., Yokelson, R. J., Emmons, L. K., Al-Saadi, J. A., Orlando, J. J., & Soja, A. J. (2011). The Fire INventory from NCAR (FINN): A high resolution global model to estimate the emissions from open burning. *Geoscientific Model Development*, 4(3), 625–641. <https://doi.org/10.5194/gmd-4-625-2011>